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Course Info

- Thank you for joining, and I hope to get a 5-star rating from you
- Covers the four domains listed in the PMI-ACP exam content outline published by PMI
- Complete all videos and practice exams at the end of each section. Try to get at least 80% and above.
 - It is not necessary to pass the exams to get the certificate, but you must do them.
- Review the exam review section at the end of the course to reinforce what you have learned
- Submit a request for a certificate of completion for the 28 PDUs required to submit your application



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Eligibility Requirements

- Secondary degree, such as a high school diploma, GED (general educational development), or global equivalent
- 24 months/ 2 years of agile experience within the last 5 years (a current PMP® satisfies this requirement but is not mandatory)
- 28 hours of training in agile practices, frameworks, and methodologies (This Course)



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PMI-ACP Exam Info

- 120 multiple choice questions (only 100 will be scored)
- 180 Minutes
 - 1.5 Minutes per Question (90 Seconds)

180 Minutes

Questions 1-60

10-Minute Break

90 Minutes

Questions 61-120



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PMI-ACP Exam Info

- Can be taken online or in-person testing center
 - Testing Center:
 - Good if you have unreliable internet or computer
 - Takes longer to get exam dates
- Online exams:
 - Easier to get exam dates
 - Less stressful
 - Computer Will need:
 - Reliable Internet
 - Camera
 - Mic
 - Speakers



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PMI Audits

- About 10% of students are audited by PMI
- Will have to mail them the following:
 - Copy of this course certificate
 - Signed experience forms
- Complete your application like you are being audited



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EXAM CONTENT OUTLINE

Domain	Percentage
Domain I. Mindset	28%
Domain II. Leadership	25%
Domain III. Product	19%
Domain IV. Delivery	28%
Total	100%



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Tips for Success

- Start working on your application right away
- Try to fill in parts of it as the course is progressing
- Make a study schedule that is realistic and stick to it
- Try to complete this course in 2-4 weeks max
- Submit your application as soon as you are done with the class
- Pre-schedule your exam as soon as you get an approval from PMI



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PM Basics



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Definition: Project

- A temporary endeavor that produces a unique product, service, or result
- Temporary in nature and has a definite beginning and ending
- Can be part of large Program or Portfolio





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Project Management

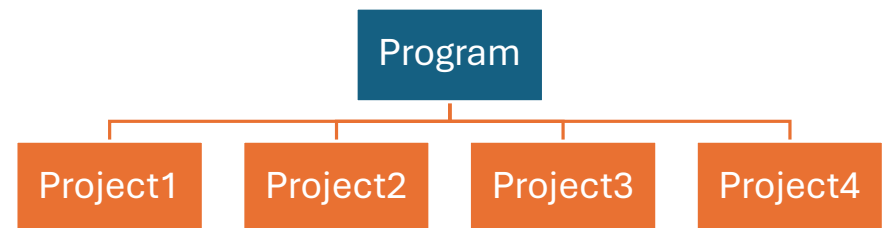
- Project management is the application of knowledge, skills, tools, and techniques to satisfy project requirements
 - Preparing a business case to justify the investment
 - Estimating resources and times
 - Developing and implementing a management plan for the project
 - Leading and motivating the project delivery team
 - Managing the risks, issues, and changes on the project
 - Monitoring progress against plan
 - Closing the project in a controlled fashion when appropriate



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Program Management

- Group of related projects managed in a coordinated way to obtain benefits and control not available from managing them individually
 - Must be some value add in managing them together as a program
 - A project may or may not be part of a program, but a program will always have projects
 - Focuses on the project interdependencies and helps to determine the optimal approach for managing them

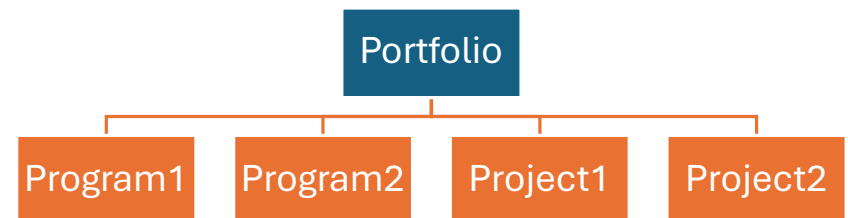




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Portfolio Management

- A portfolio is a collection of projects, programs, subsidiary portfolios, and operations managed as a group to achieve strategic objectives.
 - Collections of Projects, Programs, subsidiary Portfolios
 - Achieve strategic (long term) objectives





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Operations Management

- Deals with the ongoing production of goods and/or Services
- Considers the acquisition, development, and utilization of resources that firms need to deliver the goods and services



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Value of Projects

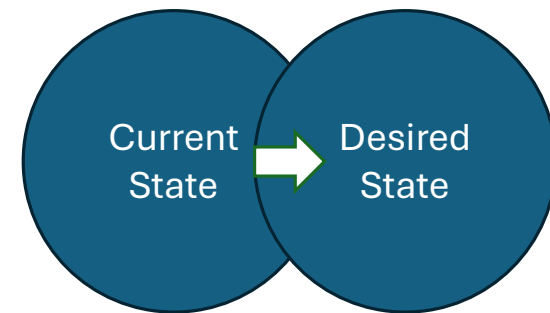
- What value will this project bring to the company upon completion?
- Why should we undertake this project?
- Money, Brand Reputation, Customer Service, New or Change product or Service



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Projects enable changes

- Project can be a vehicle for change in an organization
- Takes a company from a current state to desired state

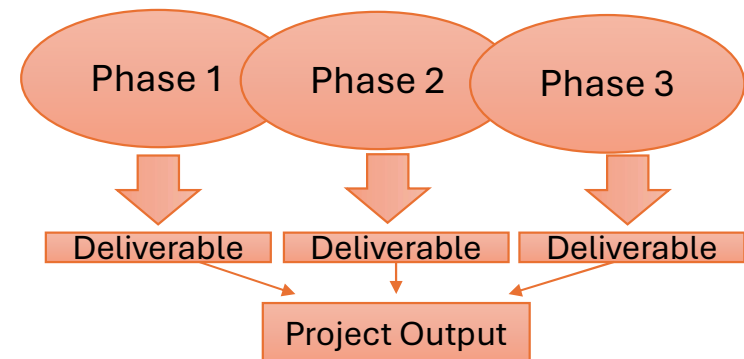


current state vs desired
state



Phases and Deliverables

- A phase is a collection of logically related project activities that culminates in the completion of one or more deliverables.
 - The number of the phases depends on the industry type and size and the complexity of the project
- A deliverable is any unique and verifiable product, service or result.
 - May be tangible or intangible
 - Must be accepted by the customer or sponsor for the phase

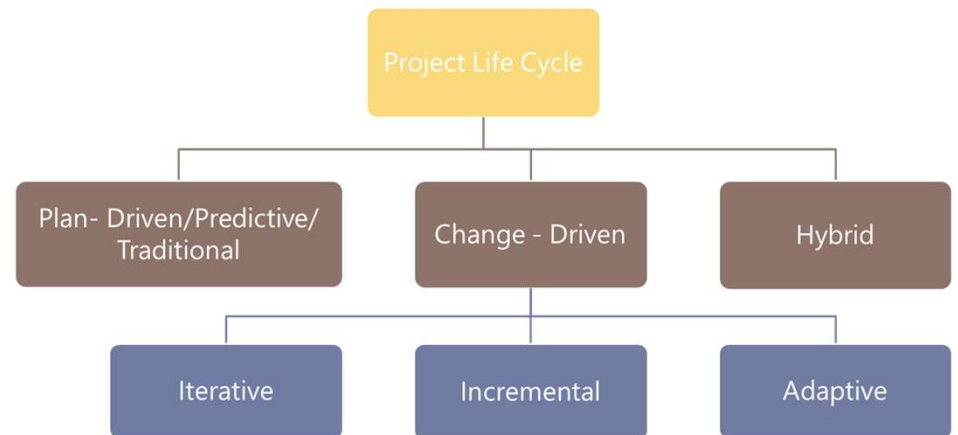




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Project Life Cycle

- A representation of the phases that a project typically goes through from start to finish
- Can be either predictive or adaptive





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Project Governance

- **Project Governance**

- Framework within which project decisions are made
- Three pillars:
 - Structure
 - People
 - Information



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Stakeholders

- Individuals, group, or organization that may affect, be affected, or perceive to be affected by the project.

Key Stakeholders

- **Project Manager** - manages the project
- **Customer** - uses the project deliverable
- **Project team** - the collection of individuals completing the project work
- **Project Sponsor** – Provides resources and support
- **Functional Manager** - Departmental Manager, i.e.: Manager of Engineering, Vice President of Marketing, Director of IT. Generally controls resources



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Project Manger Role

1. **Initiator:** An initiator is someone who takes the first step or proposes an idea or action. They are proactive in bringing forth new concepts, projects, or solutions to problems. Initiators are often innovative, creative, and forward-thinking individuals who drive the initiation and development of ideas within a group or organization.
2. **Negotiator:** A negotiator is an individual who engages in discussions or formal negotiations to reach mutually agreeable solutions or outcomes. Negotiators focus on finding common ground, resolving conflicts, and reaching compromises that satisfy the interests of all parties involved. They possess strong communication, problem-solving, and interpersonal skills to facilitate effective negotiation processes.
3. **Listener:** A listener is someone who pays attention to others' ideas, thoughts, or concerns during conversations or discussions. They give importance to active listening, empathizing with speakers, and comprehending the messages being conveyed. Listeners create a supportive and respectful environment, encouraging open communication and fostering understanding among team members.
4. **Coach:** A coach is a person who guides and supports individuals or teams in achieving their goals, improving performance, and enhancing their skills. Coaches provide constructive feedback, offer guidance, and assist in developing action plans. They help individuals unlock their potential, overcome challenges, and maximize their personal and professional growth.
5. **Working member:** A working member refers to an individual who actively participates as a member of a team or group. They contribute their knowledge, skills, and efforts towards achieving collective objectives. Working members collaborate with others, fulfill assigned tasks, and take responsibility for their contributions within the team dynamics.
6. **Facilitator:** A facilitator is someone who manages group processes, discussions, or meetings to ensure effective communication, collaboration, and decision-making. They create a structured environment, encourage equal participation, and help the group achieve its intended outcomes. Facilitators remain neutral, unbiased, and focus on maintaining a productive and inclusive atmosphere.



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Project Manger Role

- **Initiator:** An initiator is someone who takes the first step or proposes an idea or action.
- **Negotiator:** A negotiator is an individual who engages in discussions or formal negotiations to reach mutually agreeable solutions or outcomes.
- **Listener:** A listener is someone who pays attention to others' ideas, thoughts, or concerns during conversations or discussions.



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Project Manger Role

- **Coach:** A coach is a person who guides and supports individuals or teams in achieving their goals, improving performance, and enhancing their skills.
- **Working member:** A working member refers to an individual who actively participates as a member of a team or group.
- **Facilitator:** A facilitator is someone who manages group processes, discussions, or meetings to ensure effective communication, collaboration, and decision-making.



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Milestone vs task duration

- **Milestone:** A milestone is a significant event or achievement in a project that marks a key point or completion of a major deliverable.
- **Task Duration:** Task duration refers to the amount of time it takes to complete a specific task or activity within a project.

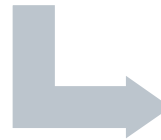


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Project Roles

**Project
Manager**

- Empowered to lead the project
- Authorized to make decisions
- Responsible for the success or failure of the project



**Project
Coordinator
or**

- Weaker than the PM
- May be authorized to make decisions



**Project
Expeditor**

- Weakest role of the PM world
- Very limited decision ability



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Project Management Office

- Organizational Structure that standardizes the processes and facilitates the sharing of resources, methodologies, tools, and techniques
- **Types:**
 - **Supportive:** Supports the project manager, such as providing templates, training, or lessons learned from other projects.
 - **Controlling:** Determines the framework or methodology and use of specific forms
 - **Directive:** Controls the project. PM will be assigned and report to the PMO.



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Project Bosses

- **Sponsor**
 - Internal or External
 - Project Champion
 - Funding the project
 - Maybe used to resolve conflicts in the project
- **Program Manager**
 - Senior to Project Manager
 - Maybe responsible for several projects executing at the same time
 - Maybe used to resolve conflicts in the project



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Organizational Structures

- **Functional Organizations**

- Structure that groups staff members according to their area of expertise (sales, marketing, construction, etc.). Functional structures require the project team members to report directly to the functional manager

- **Matrix Organizations**

- There are three matrix structures: **weak**, **balanced**, and **strong**. The different structures are reflective of the project manager's authority in relation to the functional manager's authority.

- **Project Oriented Organizations, (Projectized)**

- Structure where the PM has the greatest amount of authority. The project team is assigned to the project on a full-time basis. When the project is complete, the project team members moves on to other assignments within the organization.

- **Hybrid**

- Blended type



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Organizational Structures Chart

	Functional	Weak Matrix	Balance Matrix	Strong Matrix	Projectized
PM	Little/No	Low	Low to Moderate	Moderate to High	High/Total
Resource Avail.	Little/No	Low	Low to Moderate	Moderate to High	High/Total
Budget controls	Functional Manger	Functional Manger	Mixed	PM	PM
Pm Role	PT	PT	PT/FT	FT	FT
PM Staff	PT	PT	PT/FT	FT	FT



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Project Constraints

- Scope
- Schedule
- Cost
- Risk
- Quality
- Resources





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Areas of a Project

- Scope
 - Work to be done
- Schedule
 - Time to get the work done
- Cost
 - Budget of the work
- Quality
 - Customer satisfaction of work



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Areas of a Project

- Resources
 - Managing the people and material resources
- Communications
 - All stakeholders get the correct information at the right time
- Risk
 - Identifying and responding to risk over the lifecycle of the project
- Procurements
 - Acquiring resources from outside the project team
- Stakeholders' engagement
 - Keeping all stakeholders active and alert on the project



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Project Management approaches

- Two primary approaches: predictive (also known as traditional or waterfall) and adaptive (also known as agile or iterative).
- **Predictive Approach:** The predictive approach follows a linear and sequential process.
 - It involves detailed planning and documentation upfront, with a focus on predicting and defining the project scope, objectives, timeline, and deliverables.
 - Extensive upfront planning
 - Sequential execution
 - Emphasis on control and documentation:
 - Limited flexibility, limited changes with a well define change control process



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Project Management approaches

- The **adaptive approach** focuses on flexibility, collaboration, and iterative development.
 - It embraces change throughout the project and emphasizes continuous feedback and improvement.
 - Iterative and incremental development
 - Customer collaboration
 - Emphasis on adaptability
 - Self-organizing teams
 - The adaptive approach, commonly used in agile methodologies like Scrum or Kanban



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Product vs. Project Management

- **Product Management:**

- Product management encompasses the strategic planning, development, and lifecycle management of a product or service.
- It involves understanding market needs, defining product strategy, gathering requirements, and overseeing the product's development, launch, and ongoing optimization.
- Product managers are responsible for the overall success of the product, aligning it with business objectives and customer demands.



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Product vs. Project Management

- **Project Management:**
 - Focuses on the successful execution and delivery of specific projects within a defined scope, timeline, and budget.
 - Project managers are responsible for planning, coordinating, and managing the activities required to achieve project goals.
 - They ensure the project is completed on time, within budget, and according to the specified quality standards.



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Product vs. Project Management

- Product managers work closely with project managers to translate the product strategy and roadmap into actionable project plans.
- Project managers oversee the execution of those plans, ensuring that the product is developed, tested, and delivered according to the defined specifications.
- While project management is a part of product development, it is just one component.



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Issues, risks, assumptions, and constraints

- **Risks** are potential events or situations that may occur in the future and have an impact on the project's success.
- **Issues** refer to problems or challenges that arise during the course of a project. They are typically negative events or circumstances that can hinder progress or impact project objectives.
- **Assumptions** are statements or beliefs that are considered to be true or valid for the purpose of planning and decision-making.
- **Constraints** are limitations or restrictions that affect project planning and execution.



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Leadership vs. Management

- **Management**
 - Management focuses on tasks, processes, and operations to ensure efficient execution
 - It involves planning, organizing, coordinating, and controlling resources
 - Managers have formal authority and responsibility within the organization
 - They maintain stability, control, and order, ensuring established processes and procedures are followed.
 - Focus on coordinating and directing tasks, resources, and processes to achieve predetermined objectives.



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Leadership vs. Management

- Ensure that work is completed efficiently
- Management often focuses on immediate goals and operational efficiency
- Management aims to maintain stability, implementing processes and procedures that minimize risks
- Managers focus on achieving efficiency and optimizing resources



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Leadership vs. Management

- **Leadership:**

- Focuses on inspiring and influencing others
- Involves setting direction, motivating and empowering individuals
- They inspire and influence others
- They encourage collaboration, trust, and empowerment



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Leadership vs. Management

Table 3-1. Team Management and Team Leadership Compared

Management	Leadership
Direct using positional power	Guide, influence, and collaborate using relational power
Maintain	Develop
Administrate	Innovate
Focus on systems and structure	Focus on relationships with people
Rely on control	Inspire trust
Focus on near-term goals	Focus on long-range vision
Ask how and when	Ask what and why
Focus on bottom line	Focus on the horizon
Accept status quo	Challenge status quo
Do things right	Do the right things
Focus on operational issues and problem solving	Focus on vision, alignment, motivation, and inspiration

PMI, Process Group a Practice Guide, Page:



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Emotional Intelligence

- Emotional Intelligence (EQ) refers to the ability to recognize, understand, and manage emotions, both in oneself and in others.
- It involves being aware of one's own emotions, effectively handling them, empathizing with others, and using emotions to guide thinking and behavior.
- In project management, EQ plays a significant role



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Emotional Intelligence

- **Relationship Building**
- **Communication and Conflict Management**
- **Motivation and Influence**
- **Leadership and Decision Making**
- **Stakeholders Management**



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Domain 1: Mindset



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What is Agile

- Developed for Software projects, but it is a methodology that can be used on all Projects types
- Agile is an umbrella term that is used to refer to different types of iterative development
- Scrum is the most common method of agile, there are others such as extreme programming (XP), lean development, and Kanban.



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Characteristics of four life cycles

Characteristics				
Approach	Requirements	Activities	Delivery	Goal
Predictive	Fixed	Performed once for the entire project	Single delivery	Manage cost
Iterative	Dynamic	Repeated until correct	Single delivery	Correctness of solution
Incremental	Dynamic	Performed once for a given increment	Frequent smaller deliveries	Speed

Agile leverage both aspect of Iterative and incremental

PMI Agile Practice Guide, Page 18 table 3.1

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Andrew Ramdayal – TIA Education



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Agile Benefits

- Customer involved throughout the life cycle
- Greater Customer Interaction with all stakeholders
- Constant Feedback is required to stay current and successful
- Greater Value up front
- Change is welcomed by all stakeholders



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What is Value

- "The worth, importance, or usefulness of something."
- -PMI
- In the context of **projects and project management**, value refers to the **benefits delivered in relation to the effort and cost.**
- It can be tangible (e.g., revenue, cost savings) or intangible (e.g., customer satisfaction, brand reputation), and it should align with the **organization's strategic objectives.**



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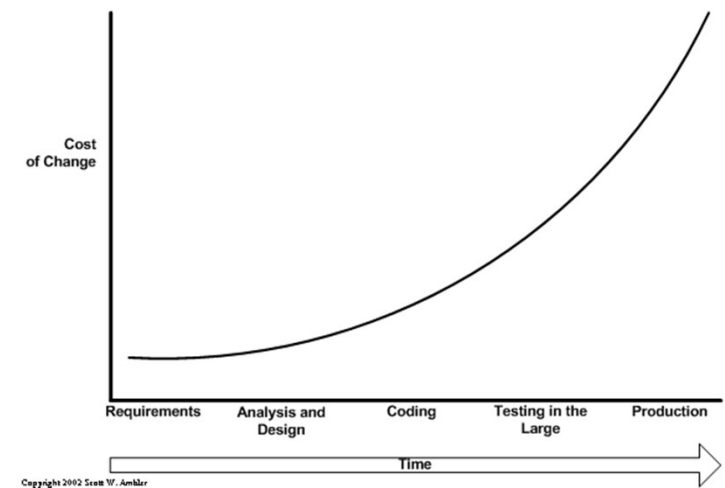
Delivering Value Incrementally

- Incremental delivery is about deploying working parts of a product over the life of the project
- In software development, its first delivered to a testing environment then to production
- This will reduce the amount of rework by discovering issues early and fixing them



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Delivering Value Incrementally



<http://www.agilemodeling.com/essays/costOfChange.htm>



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Agile ***Declaration of Interdependence (DOI)***

- *Agile and adaptive approaches for linking people, projects and value*
- *We are a community of project leaders that are highly successful at delivering results. To achieve these results:*
- *We **increase return on investment** by making continuous flow of value our focus.*
- *We **deliver reliable results** by engaging customers in frequent interactions and shared ownership.*
- *We **expect uncertainty** and manage for it through iterations, anticipation, and adaptation.*
- *We **unleash creativity and innovation** by recognizing that individuals are the ultimate source of value, and creating an environment where they can make a difference.*
- *We **boost performance** through group accountability for results and shared responsibility for team effectiveness.*
- *We **improve effectiveness and reliability** through situationally specific strategies, processes and practices.*

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Agile Mindset

- Welcoming change
- Working in small value increments
- Using build and feedback loops
- Learning through discovery
- Value-driven development
- Failing fast with learning
- Continuous delivery
- Continuous improvement



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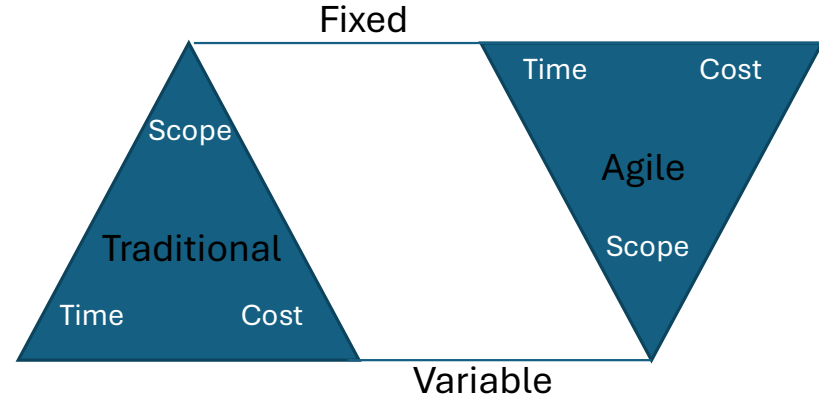
Agile vs. Traditional Project Management

- Agile builds in increments vs. as a whole
- Agile does planning throughout vs. done all at once
- Agile delivers products over time vs. all at once
- Customers sees value faster vs. at the end
- Agile wants changes vs. discouraging changes



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Inverting the Triangle





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Agile Manifesto

- Create in 2001
- Contains:
 - 4 values
 - 12 guiding principles
<https://agilemanifesto.org/>



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The Agile Manifesto Values

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

Individuals & interactions	over	Processes & tools
Working software	over	Comprehensive documentation
Customer collaboration	over	Contract negotiation
Responding to change	over	Following a plan

That is, while there is value in the items on the **right**, we value the items on the **left** more.

www.agilemanifesto.org



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Individuals and interactions over processes and tools

- While processes and tools will likely be necessary on our projects, we should focus the team's attention on the individuals and interactions involved.
- Projects are undertaken by people, not tools
- Problems get solved by people, not processes
- Projects are ultimately about people



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Working software over comprehensive documentation

- Focus on the delivering value vs. paperwork.
- Agile documents should be barely sufficient
- Done just in time
- Done just because
- Delivering software that does what it should comes first, before creating documentation.
- Agile dramatically simplify the administrative paperwork relating to time, cost, and scope control



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Customer collaboration over contract negotiation

- Be flexible and accommodating, instead of fixed and uncooperative
- Manage change, don't suppress change
- Shared definition of "done"
- Requires trusting relationship



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Responding to change over
following a plan

- Spend effort and energy
responding to changes
- Software projects tend to have
high rates of change



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Agile Guiding Principles 1-3

1. Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.
2. Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.
3. Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.



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Agile Guiding Principles 4-6

4. Business people and developers must work together daily throughout the project.
5. Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.
6. The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.



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Agile Guiding Principles 7-9

7. Working software is the primary measure of progress.
8. Agile processes promote sustainable development. The sponsors, developers)and users should be able to maintain a constant pace indefinitely.
9. Continuous attention to technical excellence and good design enhances agility.



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Agile Guiding Principles 10-12

10. Simplicity; the art of maximizing the amount of work not done is essential.
11. The best architectures, requirements and designs emerge from self-organizing teams.
12. At regular intervals, the team reflects on how to become more effective then tunes and adjusts its behavior accordingly.



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Agile Methods

- Many agile methodologies
- Scrum
- Extreme Programming (XP)
- Kanban Development
- Lean Software Development
- Scaled Agile (SAFe)



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Agile Scrum Terms

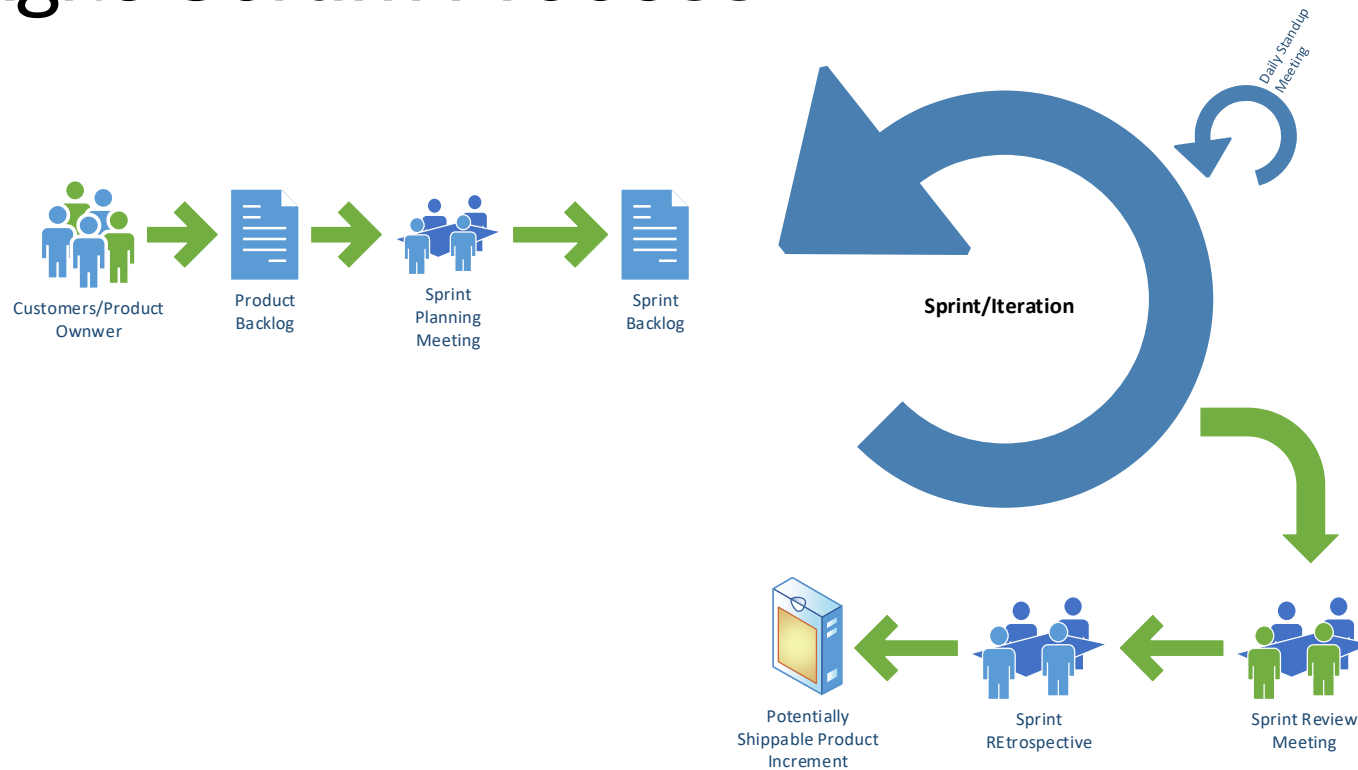
- **Product Owner** - Designated person that represents the customer on the project
- **Agile Project Manager/Scrum Master** – Manages the agile project
- **Product Backlog** - Project requirements from the stakeholders
- **Sprint Planning Meeting**- Meeting done by the agile team to determine what features will be done in the next sprint
- **Sprint Backlog** – Work the team selects to get done in the next sprint
- **Sprint** - A short iteration where the project teams work to complete the work in the sprint backlog, (1-4 weeks typical)
- **Daily Stand Up Meeting** - A quick meeting each day to discuss project statuses, led by the Scrum Master. Usually 15 minutes
- **Sprint Review** – An inspection done at the end of the sprint by the customers
- **Retrospective** – Meeting done to determine what went wrong during the sprint and what when right. Lesson learned for the sprint.
- **Partial Completed Product** - Customers Demo the product and provides feedback. This feedback adjust the next Sprint priorities
- **Release** - Several Sprints worth of work directed to operations for possible rollout and testing

Sprint = Iteration



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Agile Scrum Process





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Scrum

- Set of team guidance practices, roles, events, artifacts, and rules
- Based on three pillars of Transparency, Inspection, and Adaptation:
 - Transparency
 - Visibility to those responsible for the outcome
 - Inspection
 - Timely checks on how well a project is progressing toward goals
 - Looks for problematic deviations or differences from goals
 - Adaptation
 - Adjusting a process to minimize further issues if an inspection shows a problem or undesirable trend



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Scrum Roles & Responsibilities

Product Owner

- Owns Product vision
- Defines features, decides on release date and content
- Responsible for market success
- Prioritizes features according to market value
- Can change features and priorities every Sprint

ScrumMaster

- Responsible for facilitating process
- Focuses Team and protects them from external interruption
- Looks for ways to enhance productivity
- Assists Product Owner in leveraging Scrum



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Scrum Roles & Responsibilities

Development Team

- Small group containing all necessary project skills
- Focuses on steady delivery of high quality features
- Generates options for delivery
- Manages own work within Sprints



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Scrum Activities

The Scrum methodology refers to several different types of activities:

1. sprint planning meeting
2. sprints
 - Daily stand-up meeting
3. sprint review meeting
4. sprint retrospectives.



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Sprint Planning Meeting

- ✓ Used to determine what work will be done in that sprint and how the work will be achieved.
- ✓ The development team predicts what can be delivered based on estimates, projected capacity, and past performance to define the sprint goal.
- ✓ The development team then determines how this functionality will be built and how the team will organize to deliver the sprint goal.
- ✓ Output of this will be the sprint backlog. The work to get done in the next sprint.



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Sprints

- ✓ A sprint is a **timeboxed** (time-limited) iteration of 1-4 weeks to build a potentially releasable product
- ✓ Each sprint includes a sprint planning meeting, daily Scrum, the actual work, a sprint review meeting, and the sprint retrospective
- ✓ During the sprint, no changes are made that would affect the sprint
- ✓ The development team members are kept the same throughout the sprint



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Daily Scrum (or Standup)

- ✓ A 15-minute time-boxed activity for the Development Team to synchronize activities and create a plan for the next 24 hours
- ✓ Should be held at the same time and place each day
- ✓ Each team member should answer 3 questions:
 1. What did you do yesterday?
 2. What will you do today?
 3. Are there any impediments in your way?



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Sprint Review

- ✓ Takes place at the end of the Sprint
- ✓ Designed to gather feedback from stakeholders on what the Team has completed in the sprint
- ✓ Team demonstrates work that was completed during the sprint
- ✓ To create a conversation between the Team and the stakeholders about how to make the product better
- ✓ should be time boxed to no more than an hour per week of Sprint



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Sprint Retrospective

- ✓ Opportunity for the Team to inspect and create a plan for improvements to be done during the next Sprint.
- ✓ Team discusses:
 - ✓ What went well
 - ✓ What went wrong
 - ✓ What to do more of
 - ✓ What to do less of



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Scrum Artifacts

- ✓ Product increment
 - ✓ Part of the product that is complete after each sprint
- ✓ Product Backlog
 - ✓ Prioritized list of valuable items to deliver
- ✓ Sprint Backlog
 - ✓ List of committed items to be addressed within Sprint



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Product Backlog

- ✓ Prioritized list of all work that needs to be done to complete the product
- ✓ List is dynamics, it evolves as the more work is added and prioritized
- ✓ Items in it is prioritized by the product owner and is sorted by value
- ✓ Most valuable items are listed first
- ✓ Constantly being refined as more work is added to it.
- ✓ Team and product owner will “groom the backlog”.



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Product Increment

- ✓ Part of the product that is done after each sprint
- ✓ Done to get feedback after each sprint
- ✓ The product owner and team needs to agree upon the “definition of done” before the team starts working on the product



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Sprint Backlog

- ✓ The sprint backlog is the set of items from the product backlog that were selected for a specific sprint.
- ✓ The sprint backlog is accompanied by a plan of how to achieve the sprint goal, so it serves as the development team's forecast for the functionality that will be part of the sprint.
- ✓ It is a highly visible view of the work being undertaken and may only be updated by the development team.



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Definition of Done (DoD)

- Definition of Done (DoD) is a shared understanding of what it means when work is considered done, it should be defined at the beginning of the project, and it applies globally to the project.
- Definition of Done (DoD) is a crucial element of a successful scrum software development
- Might include things such as:
 - DoD for Unit & functional tests.
 - DoD Documentation.
 - DoD for a Writing code.



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Extreme Programming (XP)

- Software development centric agile method
- Focus software development good practices
- Scrum at the project management level focuses on prioritizing work and getting feedback



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XP Core Values

- **Simplicity**
 - Reduce complexity, extra features, and waste
 - “ Find the simplest thing that could possibly work”
- **Communication**
 - Team members know what is expected of them and what other people are working on
 - Daily stand-up meeting is key communication component
- **Feedback**
 - Get impressions of correctness early
 - Failing fast allows for faster improvement



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XP Core Values

- **Courage**
 - Allow our work to be entirely visible to others
- **Respect**
 - People work together as a team and everyone is accountable for the success or failure of the project
 - Recognize people work differently and respect those differences



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XP Roles

- **Coach**
 - Acts as a mentor, guiding the process and helping the team stay on track. Is a facilitator helping the team become effective.
- **Customer:**
 - Business representative who provides the requirements, priorities, and drives the business direction for the project.
- **Programmers**
 - Developers who build the product. Writes the codes.
- **Testers**
 - Helps the customer define and write the acceptance tests for the user stories.
Product Owner and Customer are equivalent
ScrumMaster and Coach are equivalent



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XP Practices

- Planning Activities (Games):
 - Release Planning:
 - Push of new functionality all the way to the production user
 - Customer outlines the functionality required
 - Developers estimate difficult build
 - Iteration Planning:
 - Short development cycles within a release (Scrum calls "sprints")
 - Conducted at start of every iteration, or every two weeks
 - Customer explains functionality they would like in iteration
 - Developers break functionality into tasks and estimate work
 - Based on estimates and amount of work accomplished in previous iteration,



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XP Practices

- **Small Releases:**
 - Frequent, small releases to test environments
 - Demonstrate progress and increase visibility for the customer
 - Quality is maintained: Rigorous testing or Continuous integration
- **Customer Tests:**
 - Customer describes one or more tests to show software is working
 - Team builds automated tests to prove software is working.
- **Collective Code Ownership:**
 - Any pair of developers can improve or amend any code
 - Multiple people work on all code, which results in increased visibility and knowledge of code base
 - Leads to a higher level of quality; with more people looking at the code, there is a greater chance defects will be discovered.
 - Less risk if programmer leaves, since knowledge is shared



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XP Practices

- **Code Standards:**
 - Follow consistent coding standard
 - Code looks as if it has been written by a single, knowledgeable programmer
- **Sustainable Pace:**
 - While periods of overtime might be necessary, repeated long hours of work are unsustainable and counterproductive
 - The practice of maintaining a sustainable pace of development optimizes the delivery of long-term value



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XP Practices

- **Metaphor:**
 - XP uses metaphors and similes to explain designs and create a shared technical vision.
 - These descriptions establish comparisons that all the stakeholders can understand to help explain how the system should work.
 - For example, “The invoicing module is like an Accounts receivable personnel who makes sure money collected from our customers”.
- **Continuous Integration:**
 - Integration involves bringing the code together and making sure it all compiles and works together.
 - This practice is critical, because it brings problems to the surface before more code is built on top of faulty or incompatible designs.



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XP Practices

- Test-Driven Development (TDD):
 - The team writes tests prior to developing the new code.
 - If the tests are working correctly, the initial code that is entered will fail the tests
 - The code will pass the test once it is written correctly.
- Pair Programming:
 - In XP, production code is written by two developers working as a pair to write and provide real-time reviews of the software as it emerges.
 - Working in pairs also helps spread knowledge about the system through the team.



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XP Practices

- Simple Design:
 - Code is always testable, browsable, understandable, and explainable
 - Do the simplest thing that could possibly work next. Complex design is replaced with simpler design
 - The best architectures, requirements, and designs emerge from self-organizing teams
- Refactoring:
 - Remove redundancy, eliminate unused functionality, and rejuvenate obsolete designs
 - Refactoring throughout the entire project life cycle saves time and increases quality
 - Code is kept clean and concise so it is easier to understand, modify, and extend



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Some Basic Terminology Review

Scrum	Extreme Programming (XP)	Definition
Sprint	Iteration	Fixed-length period of time (timebox)
Release	Small Release	Release to production
Sprint/Release Planning	Planning Game	Agile planning meetings
Product Owner	Customer	Business representative to project
Retrospective	Reflection	“Lessons learned”-style meeting
ScrumMaster	Coach	Agile project manager
Development Team	Team	Empowered Cross-Functional team
Daily Scrum	Daily Standup	Brief daily status meeting



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Lean Software Development

- Lean was started by Toyota as manufacturing method that was applied to software development.
- Principles:
 - Using visual management tools
 - Identifying customer-defined value
 - Building in learning and continuous improvement



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Lean Software Development





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Lean Software Development

- **Eliminate waste:**
 - To maximize value, we must minimize waste. For software systems, waste can take the form of **partially done work, delays, handoffs, unnecessary features.**
- **Empower the team:**
 - Rather than taking a micro-management approach, we should respect team member's superior knowledge of the technical steps required on the project and let them
- **Deliver fast:**
 - Quickly delivering valuable software and iterating through designs.
- **Optimize the whole:**
 - We aim to see the system as more than the sum of its parts.



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Lean Software Development

- **Build quality in:**
 - Build quality into the product and continually assure quality throughout the development process
- **Defer decisions:**
 - Balance early planning with making decisions and committing to things as late as possible.
- **Amplify learning:**
 - This concept involves facilitating communication early and often, getting feedback as soon as possible, and building on what we learn.



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Seven Wastes of Lean

1. Partially done work
2. Extra Processes
3. Extra features
4. Task switching
5. Waiting
6. Motion
7. Defects



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Kanban Development

- Kanban development is derived from the lean production system used at Toyota.
- "**Kanban**" is a Japanese word meaning "**signboard**."

Items	In Progress	Testing	Done
			
	6 cards	4 cards	



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Kanban five core principles:

- **Visualize the workflow:**
 - Software projects, by definition, manipulate knowledge, which is intangible and invisible.
- **Limit WIP:**
 - Keeping the amount of work in progress low increases the visibility of issues and bottlenecks
- **Manage flow:**
 - By tracking the flow of work through a system, issues can be identified and changes can be measured for effectiveness
- **Make process policies explicit:**
 - It is important to clearly explain how things work so the team can have open discussions about improvements
- **Improve collaboration:**
 - Through scientific measurement and experimentation, the team should collectively own and improve the processes it uses.



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Feature-Driven Development (FDD)

- An Agile and model-driven methodology
- Focuses on designing & building features (small, client-valued functions)
- Best for larger teams and complex systems
- “*Feature*” = *small, useful result valued by the client (e.g., “Calculate invoice total”)*



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Feature-Driven Development (FDD)

- The 5 Key Processes of FDD

1. Develop Overall Model
2. Build Feature List
3. Plan by Feature
4. Design by Feature
5. Build by Feature

Each process builds upon the last — it's highly structured but still Agile.



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Feature-Driven Development (FDD)

- **Develop Overall Model**
 - Understand the domain and build a high-level object model
 - Workshops with domain experts to define core concepts and relationships
 - Foundation for all future design decisions
- **Build Feature List**
 - Break down the model into a hierarchical list of features
 - Features grouped into major feature sets and subject areas
 - Each feature represents client-valued functionality
- **Plan by Feature**
 - Create a development schedule based on the feature list
 - Assign ownership of features and classes to team members
 - Allows tracking and resource planning at a granular level
- **Design by Feature**
 - Small teams design each feature in detail
 - Produce sequence diagrams and design packages
 - Led by the Chief Programmer with input from Class Owners
- **Build by Feature**
 - Code, test, and integrate the feature
 - Involves unit testing and code inspections before completion
 - Features are built in short iterations and frequently delivered



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Feature-Driven Development (FDD)

- When to Use FDD **or not**
 - Large, complex systems
 - Teams needing structure and upfront modeling
 - Projects requiring **predictability and strong design discipline**
 - Not ideal for very small teams or early-stage MVP work
 - *FDD blends Agile principles with structured engineering discipline.*



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Crystal Framework

- Crystal is a family of lightweight Agile methodologies
- Believes one size does not fit all — method adapts based on project size and criticality
- Focuses on people, interactions, community, skills, and communications
- Crystal values individuals over processes — like Agile Manifesto in action.



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Crystal Framework

- Key Principles
 - Frequent delivery of usable code
 - Reflective improvement (retrospectives)
 - Team collaboration
 - Personal safety and team trust



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Crystal Framework

Method	Team size	Project size
Crystal Clear	6 people or less	Small projects
Crystal Yellow	7–20 people	Small to medium projects
Crystal Orange	20–40 people	Medium projects
Crystal Red	40–80 people	Medium to large projects
Crystal Maroon	80–200 people	Large projects
Crystal Diamond or Crystal Sapphire	Very large teams (200+ people)	Very large projects with high criticality

<https://www.wrike.com/agile-guide/faq/what-is-agile-crystal-methodology/>



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Crystal Framework

When to Use Crystal or not

- Projects with small-to-medium teams
- When flexibility and team empowerment are needed
- Teams that prefer less documentation and more interaction
- Not ideal for highly regulated or safety-critical systems



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SAFe (Scaled Agile Framework)

- A framework for applying Agile at enterprise scale
- Combines Agile, Lean, and DevOps principles
- Designed for large organizations with multiple teams working together
- Helps enterprises deliver value faster and more efficiently at scale.



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SAFe (Scaled Agile Framework)

- Core Values:
 - Alignment
 - Built-in Quality
 - Transparency
 - Program Execution
- Principles include:
 - Apply systems thinking
 - Assume variability, preserve options
 - Build incrementally with fast, integrated learning cycles



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SAFe (Scaled Agile Framework)

When to Use SAFe or not

- Large enterprises with many Agile teams
- Need for **strategic alignment** across business and tech
- Regulatory or high-dependency environments
- May be **too heavy** for small teams or startups
- *SAFe provides structure for scaling Agile — but requires strong leadership and training.*



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Disciplined Agile (DA)

- A toolkit, not a framework — helps you choose your way of working (WoW)
- Blends Agile, Lean, Scrum, Kanban, SAFe, and even traditional methods
- Focuses on context-driven agility, not one-size-fits-all



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Disciplined Agile (DA)

DA supports multiple life cycles, including

- Agile Lifecycle (Scrum-based)
- Lean Lifecycle (Kanban-based)
- Continuous Delivery (Agile or Lean flavor)
- Program Lifecycle (large-scale projects)
- Exploratory Lifecycle (for innovation)
- Waterfall Lifecycle (for traditional teams)



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Disciplined Agile (DA)

When to Use DA or not

- Organizations seeking flexibility and guidance
- Teams that need help tailoring their WoW
- Enterprises blending multiple approaches
- Not ideal for teams wanting a prescriptive, rigid framework
- Disciplined Agile is about agility, tailor your approach for success.



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Domain 2: Leadership



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Leading Effectively

- Servant Leadership
 - Leader provides what the team needs
 1. Shield team from interruptions
 2. Remove impediments to progress
 3. (Re)Communicate project vision
 4. Carry food and water



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Twelve Principles for Leading Agile Projects

1. Learn the team members needs
2. Learn the project requirements
3. Act for the simultaneous welfare of the team and the project
4. Create an environment of functional accountability
5. Have a vision of the completed project
6. Use the project vision to drive your own behavior

©2002 Jeffery Pinto, Project Leadership from Theory to Practice



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Twelve Principles for Leading Agile Projects

7. Serve as the central figure in successful project team development
8. Recognize team conflict as a positive step
9. Manage with an eye toward ethics
10. Remember that ethics is not an afterthought, but an integral part of our thinking
11. Take time to reflect on the project
12. Develop the trick of thinking backwards



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Leadership Tools and Techniques

- Using these tools still need soft-skills approach
- Modeling Desired Behavior
 - Honesty
 - Forward-Looking
 - Competent
 - Inspiring
- Communicating project vision
- Enabling others to act
 - Switch from exclusive tools to inclusive tools
- Being willing to change the status quo



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Leadership Task

- Practice Transparency through Visualization
- Create a safe environment for experimentation
- Experiment with new techniques and processes
- Share knowledge through collaboration



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Communicating with Stakeholders

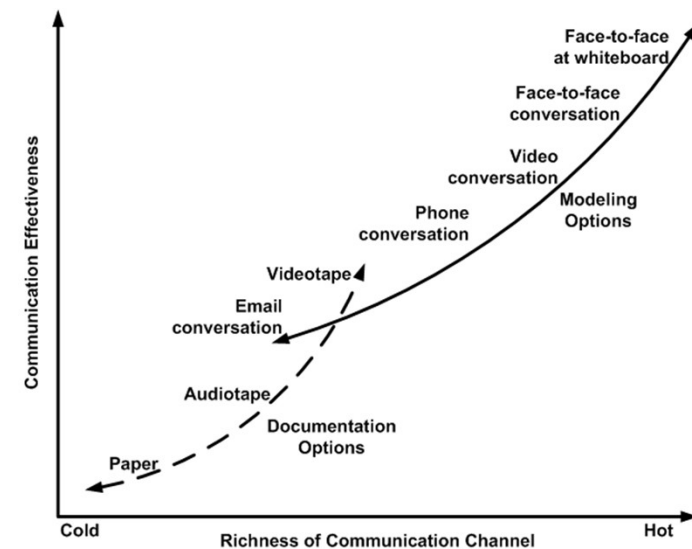
- Face to face communication
- Two-way communication
- Knowledge sharing
- Information Radiators
- Social Media



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Face-to-face Communication

- Face to face communication



Copyright 2002-2005 Scott W. Ambler
Original Diagram Copyright 2002 Alistair Cockburn

<http://www.agilemodeling.com/essays/communication.htm>



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Communicating with Stakeholders

- Two-way communication
 - Just don't ask for confirmation or concerns, but actually listen to what they have to say
- Knowledge sharing
 - Agile teams work closely with each other such as with pair-programming.
 - Using Kanban boards or wireframes are ways to share information
 - Use of low-tech tools like a whiteboard will allow all to see the work and understand it
 - We must encourage it



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Communicating with Stakeholders

- Information Radiators
 - Things that are highly visible
 - Used to display information
 - Usually includes chats, graphs and boards
- Social Media
 - Use to communicate
 - Can include twitter or Instagram



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Tools for Agile Projects

- Low-tech, high-touch over computer models
- When using computer models problems could arise such as:
 - Data accuracy perception increases
 - No stakeholder interaction. Only a few people would update them



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Low-Tech, High-Touch Tools

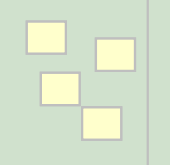
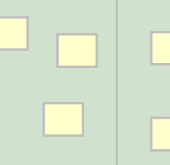
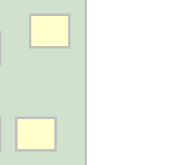
- Use card, charts, whiteboards, and walls
- Promotes communication and collaboration
- Skip using a computer Gantt chart to a Kanban board



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Kanban/Task Board

- An "information radiator" - ensures efficient diffusion of information
- Can be drawn on a whiteboard or even a section of wall
- Makes iteration backlog visible
- Serves as a focal point for the daily meeting

Items	In Progress	Testing	Done
			
	6 cards	4 cards	



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Educating People about Agile

- Teach all the stakeholders about the benefits of agile
- Concerns about agile can Include:
 - Senior management and sponsor: They are worried about the risk of failing
 - Managers: fear the loss of control
 - Project team: resist agile methods
 - Users: will not get all features



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Engaging Stakeholders

- Short iterations and release keeps them engage
- Keeping them engage can lead to stakeholders being more involved and getting more change request
- This helps us to identify risk and issues early
- If some stakeholders are causing problems, the agile PM will need to use their interpersonal skills to resolve issues
- Need to have a process for escalating stakeholders issues
- Why such a big focus on stakeholders?
 - Projects and done by people for people



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Methods of Stakeholder Engagement

- Get the right stakeholders
- Cement stakeholder involvement
- Actively manage stakeholder interest
- Frequently discuss what done looks like
- Show progress and capabilities
- Candidly discuss estimates and projections



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Set a Shared Vision

- Important to ensure customers and agile project team has the same vision
- Methods include:
 - Agile Charter
 - Definition of “Done”
 - Agile Modeling
 - Use case diagram
 - Data models
 - Screen design
 - Wireframes
 - Personas



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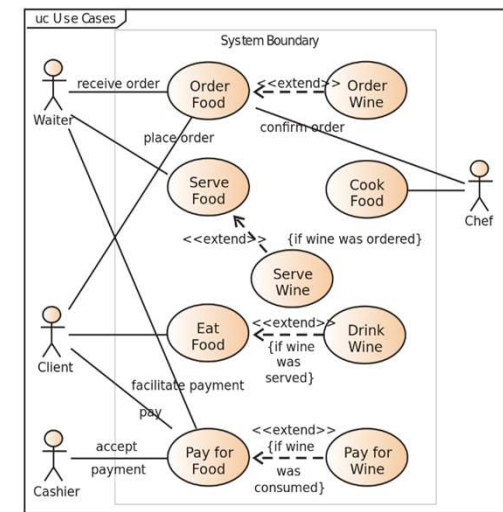
Definition of “Done”

- Creating a shared vision of what done looks like
- Should be done for:
 - User stories
 - Releases
 - Final project deliverables



Agile Modeling

- Use case diagrams
 - Visually shows how users would use an application

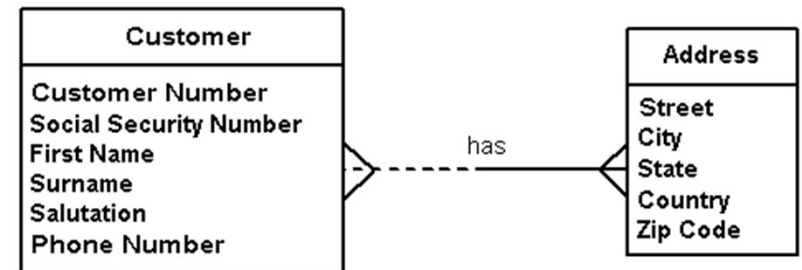


https://en.wikipedia.org/wiki/Use_case_diagram



Agile Modeling

- Data models
 - How the data are structured in tables and their relationships



Copyright 2002-2006 Scott W. Ambler

<http://www.agiledata.org/essays/dataModeling101.html>



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Agile Modeling

- Screen designs
- Simple screen shots

The image shows two hand-drawn screen designs. The left screen is titled 'Student Information' and contains fields for Student Number, First Name, Middle, Surname, Salutation, and Date First Enrolled. Below these is a table for 'Seminar' with columns for Seminar, Term, Age, and Status. The right screen is titled 'Add a seminar' and contains a form for Seminar Name, a 'Search' button, and a 'Results' table with columns for Seminar, Term, Seats, and Duration. Below the table is a 'Course description' section.

Student Information

Student Number: 789-587-234

First Name: Scott

Middle: E. Clinton

Surname: Johnson

Salutation: Mr.

Date First Enrolled: June 11 2003

Seminar:

Seminar	Term	Age	Status
CSC 100 Intro to CS	Fall 2003	A+	Passed
CSC 200 Intro to AI	Fall 2003	A	Passed
CSC 300 Advanced AI	Spring 2004	-	Enrolled

Add a seminar

Seminar Name: [CSC #]
[Search]

Results

Seminar	Term	Seats	Duration
CSC 350 Agile Techniques	Fall 2003	4	Scrum
CSC 300 Agile Exp	Spring 2003	17	Scrum
CSC 310 Agile Decision Techniques	Spring 2004	8	Scrum

Course description:
CSC 300 Agile Decision Techniques
This course describes evolutionary development strategies for data oriented development. See www.agilemodeling.org for details.
This course currently has 39 people waitlisted for it.

<http://agilemodeling.com/artifacts/uiPrototype.htm>



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Personas

- Personas
 - Quick guides or reminders of key stakeholders and interests
 - Provide description of users
 - Be grounded in reality
 - Be goal-oriented, specific, and relevant
 - Be tangible and actionable
 - Generate focus
 - Help team focus on valuable features to users



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Personas

Name: Andrew Jones– Certified Accountant

**Description:**

Andrew has been an Accountant for over 10 years and has worked at many large accounting firms.

He likes to be organized and get his work done on time.

Value:

Andrew would like to ensure all company bills are paid on time while using online auto payments.

He would like to ensure customers are reminded automatically of outstanding balances.

He is looking to print the receivables and payable reports on a weekly basis to check on bills and invoices.



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Development / Delivery Team

- Group that build and test the increments of the product
 - Build product in increments
 - Update information radiators
 - Self organize and directing
 - Share progress by doing daily stand-up meetings
 - Demo the completed product increments
 - Holds retrospectives at the end of sprints
 - Does release and sprint planning and estimations



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Product Owner/Customer

- Prioritizing the product features
- Manage the product backlog ensuring its accurate and up to date
- Ensures the team has a shared understanding of the backlog items
- Defines the acceptance criteria
- Provides the due dates for the releases
- Attends planning meeting, reviews, and the retrospective.



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Agile Project Manager (ScrumMaster/Coach)

- Act as a servant leader
- Help the team self-organize and direct themselves
- Be a facilitator
- Ensure the team plan is visible and the progress is known to the stakeholders
- Act as a mentor and coach
- Work with the product owner to manage the product backlog
- Facilitates meeting
- Ensure issues are solved

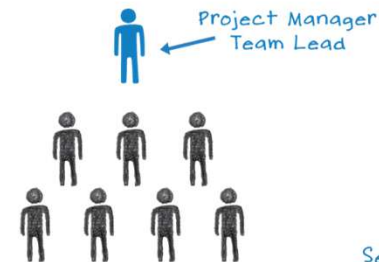


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Building Teams

- Self-Organizing
- Self-Directing
- Small teams with fewer than 12 members

Traditional Teams



Agile Teams





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Generalizing Specialists

- Have members that can do different tasks
- Members skilled in more than one area
- Share work reduce bottleneck



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High-Performance Agile Teams

- Have a shared vision
- Realist goals
- Fewer than 12 members
- Have a sense of team identity
- Provide strong leadership



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Experiments (Have a safe place)

- Establish safe environment for disagreement
- Allows team members to build strong commitment to decisions
- Encourage people to experiments with new methods
- Leads to more engagement



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Welcome Constructive Disagreement

- Leads to better buy-in and decisions
- Avoiding conflicts can lead to conflicts escalating
- A safe place for disagreement leads to successful problem solving



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Team Spaces

- Co-located Teams
- Team Spaces
- Osmotic Communication
- Global and Cultural Diversity
- Distributed teams



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Co-Located Teams

- All team member work together in the same location
- Allows for face-to-face time and interaction
- Should be within 33 feet of each other
- No physical barriers
- Sometimes a virtual co-location



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Team Space

- Lots of low-tech, high touch
 - Whiteboards and task boards
 - Sticky notes, flip charts
 - Round table
 - No barriers to face-to-face communication
- Caves and Common
 - Caves → space team members can retreat to individually
 - Common → space team members can work as group
- Osmotic Communication
 - Information flows that occur as part of everyday conversations and questions
 - 33 feet or 10 meters
- Tacit Knowledge
 - Information that is not written down; supported through collective group knowledge



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Global and Cultural Diversity

- Time Zones
- Cultures
- Native Languages
- Styles of
communications



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Distributed Teams

- At least one team member working off-site
- Need to find ways to replicate co-location team benefits
- Agile Tools
 - Low-Tech, High-Touch Tools
 - Digital Tools for distribute teams
 - Video conferencing
 - Interactive whiteboards
 - IM / VoIP
 - Virtual card walls
 - Web cams
 - Digital cams



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Models of team development

- Shu-Ha-Ri Model of Skill Mastery
 - Shu- Obey,
 - Ha – Moving away,
 - Ri – finding individual paths
- Dreyfus Model of Adult Skill Acquisition
 - Novice, Advanced Beginner, Competent, Proficient, Expert



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Tuckman's Five Stages of Team Development

1. **Forming:** team comes together and starts to get to know each other. There is not much conflict or communication.
2. **Storming:** team members start to have conflicts with each other. They start to learn of each other's ideas and may not agree with them. Most conflicts takes place in this stage.
3. **Norming:** the team members begin to agree with each other on the best methods to build the deliverables. Generally, everyone is coming to a consensus.
4. **Performing:** the team is performing well and is working without conflict.
5. **Adjourning:** In this stage, the project is completed and the team is reassigned.



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Solving Problems

- Problem Solving as continuous improvement
- Engage the team
- Some problems can't be solved



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Why Engaging the Team?

- Team usually produces the best practical solutions
- Benefits
 - Get consensus from all members
 - Gets a broad knowledge base
 - Solutions are practical
 - When ask people work hard to produce good ideas
 - Asking someone for help shows confidence
- Usage and Cautions
 - Solve real problems
 - Poor team cohesion
 - Team and project changes
 - Follow-Through



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Verifying and Validating Value

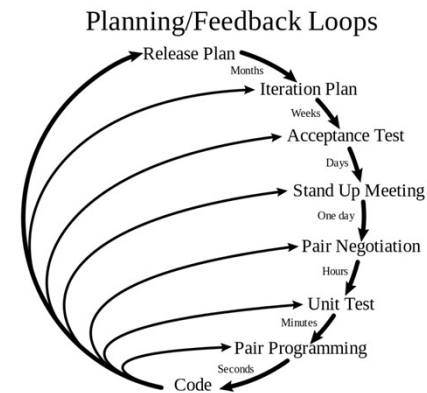
- “Gulf of Evaluation”
 - What one person describes is often different from how another interprets



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Frequent Verification and Validation

- Resolve problems as soon as possible
- Don't let little problems grow over time





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Negotiation

- This happens all throughout the project
- Good negotiation will allow everyone to investigate the options and trade-offs
- Most effective when interactions between people are positive and there are room for give and take



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Using Workshops

- Meeting when work gets done
- Retrospectives are a type of workshops
- Ways to make them more effective:
 - Diverse groups has a larger perspective
 - Use methods such as round-robin to ensure no one dominates
 - Try to get everyone to participate in the first few minutes
- User story workshops are where we write the user stories and keep stakeholders engage



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Active Listening

- Level 1: Internal – how is it going to affect me
- Level 2: Focused – put ourselves in the mind of the speaker
- Level 3: Global – builds on level with body language



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Brainstorming

- Brainstorming
 - Quiet Writing
 - Give people about 5 minutes to write down their ideas
 - Round-Robin
 - Pass a token around to ensure everyone will speak
 - Free-for-all
 - People shout out their thoughts. May only work in a supportive environment



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Facilitation

- Run effective meeting and workshops.
- Have the following:
 - Goals
 - Rules
 - Timing
 - Assisting



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Failure Modes

Why do people Fail:

1. Making mistakes
2. Preferring to fail conservatively
3. Inventing rather than researching
4. Being creatures of habit
5. Being inconsistent



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Success Modes

- Why do we succeed:
 1. Being good at looking around
 2. Being able to learn
 3. Being malleable
 4. Taking pride in work



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Success Strategies

- Balance discipline with tolerance
- Start with something concrete and tangible
- Copy and alter
- Watch and listen
- Support both concentration and communication
- Match work assignment with the person
- Retain the best talent
- Use rewards that preserve joy and combine rewards
- Get feedback



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Green Zone/Red Zone

- Red Zone:
 - Blames others for everything
 - Responds defensively
 - Feels threatened
 - Triggers defensiveness
 - Doesn't let go or forgive
 - Uses shame and blame
 - Focus on short-term advantage
 - Doesn't seek or value feedback
 - Sees conflict as a battle and only seeks to win
 - Communicates high level of disapproval
 - Sees others as the problem or enemy
 - Does not listen effectively



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Green Zone/Red Zone

Green Zone:

- Take responsibility
- Seeks to respond nondefensively
- Is not easily threatened psychologically
- Attempts to build success
- Uses persuasion rather than force
- Thinks both short and long term
- Welcomes feedback
- Sees conflict as a natural part of life
- Seeks excellence rather than victory
- Listens well



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Conflict Resolution

- All projects will have conflicts
- While some level of conflicts are good, we need to ensure they don't become a "world war" where people are trying to destroy each other
- Levels of conflict(1-5):
 - Level 1: Problem to solve – sharing info
 - Level 2: Disagreement – Personal Protection
 - Level 3: Contest – Must win
 - Level 4: Crusade – Protecting one's group
 - Level 5: World War – Must destroy the other



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Participatory Decision Models

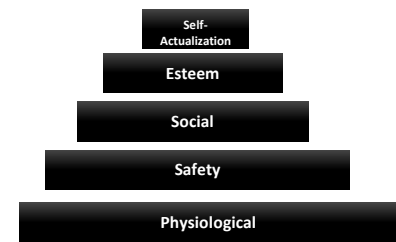
- Engage stakeholder in decision making process
 - Simple voting
 - Vote “for” or “against” it
 - Thumps up/down/sideways
 - People hold their thumps in a way of if they support it or not. Sideway is if they cannot make up their mind
 - Fist of five
 - People show up finger based on they support the idea
 - 1 finger: total support – 5 finger: Stop against it



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Theory of Motivation

- Recognition and Rewards
 - Rewarding good behavior, Only desirable behavior should be rewarded, used to increase morale
- Maslow's hierarchy of needs
 1. Physiological: The necessities to live: air, water, food, clothing, and shelter.
 2. Safety: People need safety and security; this can include stability in life, work, and culture.
 3. Social: People are social creatures and need love, approval, and friends.
 4. Esteem: People strive for the respect, appreciation, and approval of others.
 5. Self-actualization: At the pinnacle of needs, people seek personal growth, knowledge, and fulfillment.





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Theory of Motivation

- Herzberg's Theory of Motivation
 - Hygiene agents (What factors influence satisfaction at work) are expected by and can only demotivate if they are not present. Motivating agents provide opportunity to exceed, and advance.
- McGregor's Theory X and Y
 - Theory X- is bad. These people need to be watched all the time, micromanaged, and distrusted, people avoid work, responsibility, and have no ability to achieve.
 - Theory Y is good. These people are self-led, motivated, and can accomplish new tasks proactively.
- Theory Z
 - Increased Loyalty at the workplace. Theory emphasizes the well-being of the employees, both at work and outside of work, it encourages steady employment



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Theory of Motivation

- Expectancy Theory
 - People behave based on what they expect as a result of their behavior.
- McClelland 3 need theory
 - Achievement
 - Power
 - Affiliation
- Forms of Power
 - Reward Power - Ability to give rewards
 - Expert Power - SME
 - Legitimate(formal power)
 - Referent- Respect /Personality of the Manger
 - Punishment- Punish associates when they fail (least desirable)



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Domain 3: Product



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Early Value Delivery

- Agile promote early and often delivery
- Aim to deliver highest value early in project
 - Deliver as many high-value components as soon as possible
 - Stakeholder satisfaction → Project success
 - Shows understanding of stakeholders' needs
 - Stakeholders are engaged
 - Builds confidence of stakeholders in time



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Adaptive Planning

- Planning is ongoing process
- Multiple mechanisms to proactively update plan
- Focus on value delivery and minimize nonvalue-adding work
- Uncertainty drives need to replan
- Frequently discover issues and experience high rates of change



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Agile Plans

- Agile planning varies from traditional planning
 1. Trial and demonstration uncover true requirements, which then require replanning
 2. Agile planning is less of an upfront effort, and instead is done more throughout the project
 3. Midcourse adjustments are the norm



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Principles of Agile Planning

1. Plan at multiple levels
2. Engage the team and the customer in planning
3. Manage expectations by frequently demonstrating progress
4. Tailor processes to the project's characteristics
5. Update the plan based on the project priorities
6. Ensure encompassing estimates that account for risk, distractions, and team availability
7. Use appropriate estimate ranges to reflect the level of uncertainty in the estimate
8. Base projections on completion rates
9. Factor in diversion and outside work



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Value-Driven Delivery

- Projects undertaken to generate business value
 - Produce Benefit
 - Improve Service
 - Market Demand
 - Safety Compliance
 - Regulatory Compliance
 - Increase Revenue
 - Decrease Expense

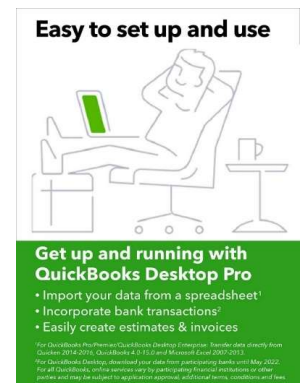
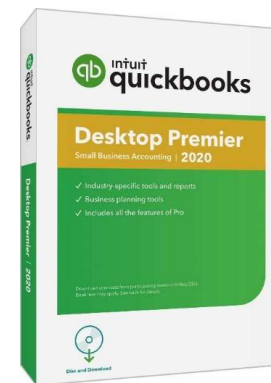


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Value-Base Analysis and Decomposition

- Assessing and prioritizing the business value of work items, and then plan accordingly.
- Value-Based Decomposition
 - Breaks down requirements and prioritized them
 - Design the product box

Design the Product Box





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Agile Chartering

- High-level (uses the W5H)
- Agreement
- Authority to proceed
- Focuses on *how* project will be conducted
 - Allows for flexibility and ability to deal with change
- Project specific processes outlined
- May use project Tweet – Describes project goal in 140 Characters or less.



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Regulatory Compliance

- Mandated requirements usually by government agencies
- Must be implement into the project work as regular work



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People Over Processes

- Projects are done by people, not tools
 - Agile manifesto: “Individuals and Interactions over processes and tools”
- Focus on the people side of the project
- Projects are more about people management than tools management



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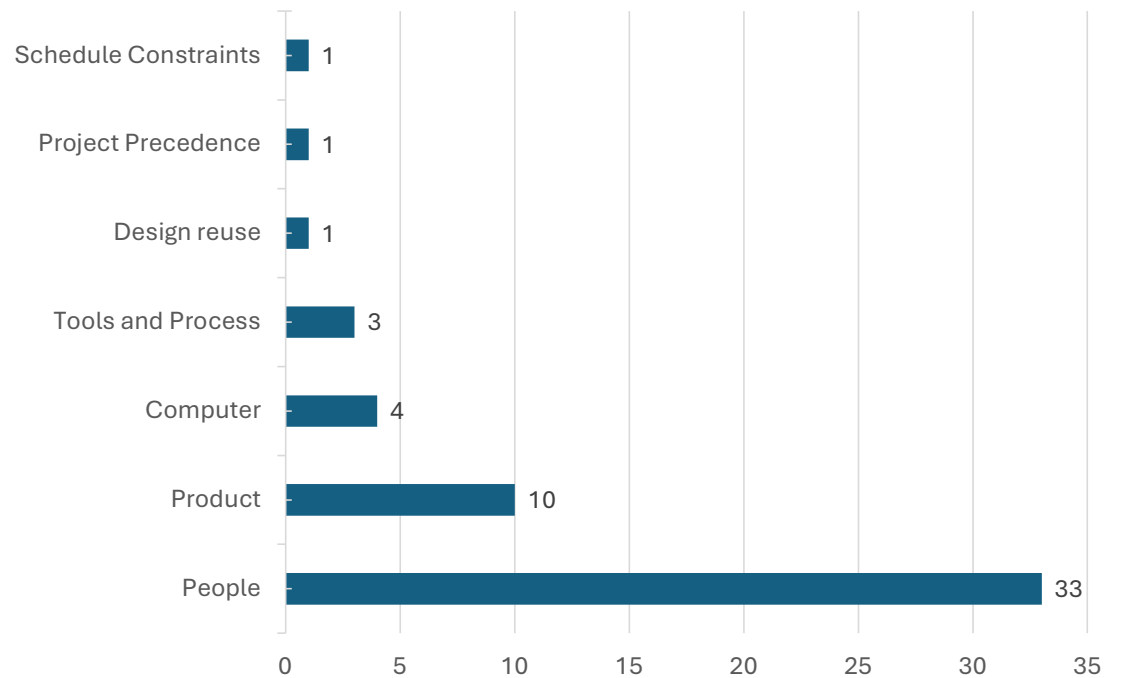
People Over Processes

- COCOMO
 - Constructive Cost Model
 - To determine correlation between project input variables and final cost to use to estimate future projects
 - People factors has a score of 33...11 times more significant than tools and processes



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COCOMO II

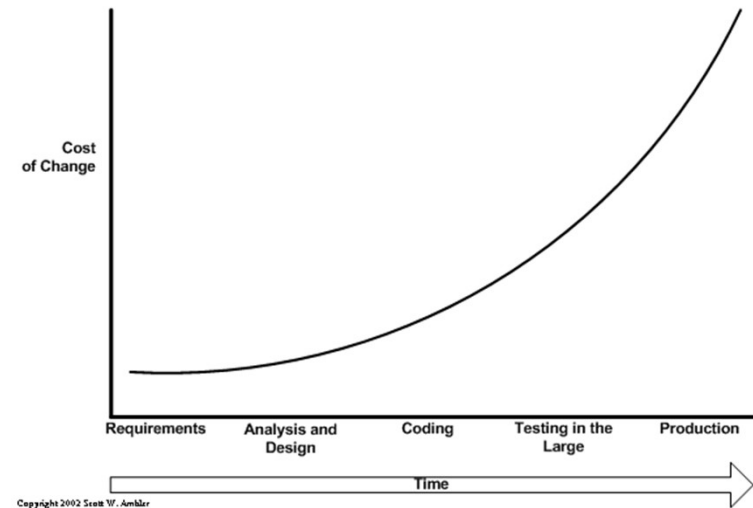




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Cost of Change

- Over time the cost of change will increase





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Assessing Value - Financial Metrics

- Return on investment (ROI)
 - The ratio of the benefits received from an investment to the money invested. Usually a percentage
- Internal rate of return (IRR)
 - Interest rate you will need to get in today's money to receive a certain amount of money in the future
- Present Value/Net Present value (NPV)
 - Present Value (PV) is the current worth of a future amount of money, discounted using a specific interest rate.
 - Net Present Value (NPV) is the total present value of all expected future cash inflows minus the initial investment cost—used to determine if a project is financially viable.



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Assessing Value - Financial Metrics

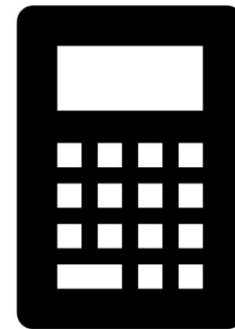
- Earned Value Management
 - Formulas that monitor the value of the project as its progressing.



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PMP/CAPM Formulas

- Earn Value Management (EVM)
- Program Evaluation and Review Technique (PERT)
- Communication Channels





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PMP Exam

- 2-5 Questions
- **Mostly Likely** will not need to calculate any formulas
- Will need to interpret the results
- **Don't necessarily** need to memorized the formulas



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CAPM Exam

- 4-8 Questions
- **Will need** to calculate formulas
- Will need to interpret the results
- **Memorized the formulas**



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Earned Value Management

- Series of formulas used in cost management
- Used to give status about the project work.
- Answers questions such as:
 - Is the project on budget
 - Is the project on schedule
 - What will this project cost when completed (forecast)



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Earned Value Management

- **BAC** = Total Budget Cost for the project (No Formula)
- **PV** = Planned % Complete $\times$ **BAC**
- **EV** = Actual % $\times$ **BAC**
- **AC** = Sum of the cost for the given period (No Formula)
- **CV** = **EV** - **AC**
- **CPI** = **EV** / **AC**
- **SV** = **EV** - **PV**
- **SPI** = **EV** / **PV**
- **EAC** = **BAC** / **CPI**
- **ETC** = **EAC** - **AC**
- **VAC** = **BAC** - **EAC**
- **TCPI** = $(\text{BAC} - \text{EV}) / (\text{EAC} - \text{AC})$



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EVM Formula: BAC

- **Term:** Budget at Completion (BAC)
- **Description:** Original budget of the project.
- **Formula:** None, just the original budget.



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EVM Formula: PV

- **Term:** Planned Value (PV)
- **Description:** Amount of money worth of work we that should have been done on the project.
- **Formula:**
 - $PV = \text{Planned \% Complete} \times BAC$



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EVM Formula: EV

- **Term:** Earned Value (EV)
- **Description:** Amount of money worth of work you actually did on the project.
- **Formula:**
 - $EV = \text{Actual \% Complete} \times \text{BAC}$



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EVM Formula: AC

- **Term:** Actual Cost (AC)
- **Description:** Amount of money you already spent on the project
- **Formula:**
 - None, just the amount already spent on the project.



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EVM Formula: CV

- **Term:** Cost Variance
- **Description:** The difference between the work done and money spent. This value should be positive for under budget. Negative values indicate over budget
- **Formula:**
 - **$CV = EV - AC$**



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EVM Formula: CPI

- **Term:** Cost Performance Index
- **Description:** The rate of how we are spending to actually earning on the project. **This value should be 1 and over for projects under budget.**
- **Formula:**
 - $CPI = EV / AC$



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EVM Formula: SV

- **Term:** Schedule Variance (SV)
- **Description:** The difference between the amount of work we should have done vs. the amount actually done. **This value should be positive for ahead of schedule. Negative values indicate behind schedule**
- **Formula:**
 - $SV = EV - PV$



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EVM Formula: SPI

- **Term:** Schedule Performance Index (SPI)
- **Description:** The rate of how we are meeting the project schedule. **This value should be 1 and over for a project to be ahead of the schedule.**
- **Formula:**
 - $SPI = EV / PV$



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EVM Formula: EAC

- **Term:** Estimate at Completion (EAC)
- **Description:** Forecasting the total cost of the project at the end based on the current spending rate of the project.
- **Formula:**
 - $EAC = BAC / CPI$



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EVM Formula: ETC

- **Term:** Estimate to Completion (ETC)
- **Description:** Forecasting the amount that will be needed to complete the current project based on the current performance
- **Formula:**
 - $ETC = EAC - AC$



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EVM Formula: VAC

- **Term:** Variance at Completion (VAC)
- **Description:** The difference between the original budget and new forecasted budget. This value should be positive for projects that may end at or under budget
- **Formula:**
 - $VAC = BAC - EAC$



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EVM Formula: TCPI

- **Term:** To-Complete Performance Index (TCPI)
- **Description:** The performance that needs to be met to finish the project within the budget
- **Formula:**
 - $TCPI = (BAC - EV) / (BAC - AC)$



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3 Point Formula: PERT (Beta)

- **Description:** Estimates time or cost with more weight on realistic.
- **Formula:**
 - $(\text{Optimistic} + (4 \times \text{Realistic}) + \text{Pessimistic}) / 6$



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3 Point Formula: PERT (Tri)

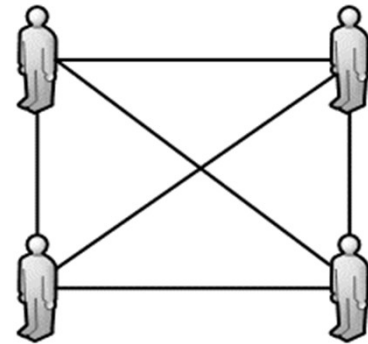
- **Description:** Estimates time or cost with evenly across the 3 values.
- **Formula:**
 - **(Optimistic + Realistic + Pessimistic)/3**



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Communication Formula

- **Description:**
Communications
Channels
- **Formula:**
 - **Channels = $n(n-1)/2$**





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Minimal Viable Product (MVP)

- Refers to a set of functionality that is complete to be useful, but small enough not to be an entire project
- Usually a module in a software



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Minimal Marketable Feature

- **MMF:**
 - The smallest set of functionality that delivers real value to users and can be released independently
 - Must be usable, testable, and valuable
 - Helps deliver working software early and often
 - Reduces time-to-market and enables faster feedback
 - Not just a feature — it must be marketable (usable by the customer or user)



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MMF vs MVP

Minimum Viable Product (MVP)

- Built to test an idea or assumption
- Goal: Learn fast with minimal effort
- May not be polished — just “viable” enough to gather feedback
- Often used in early product development or startup experiments
- Example: A basic web page to validate interest in a new service

Minimal Marketable Feature (MMF)

- A complete, deliverable feature that provides real value to users
- Ready for release to customers
- Has business value and can be monetized
- Used in Agile to ship value early and iteratively
- Example: A fully working checkout system in an e-commerce app



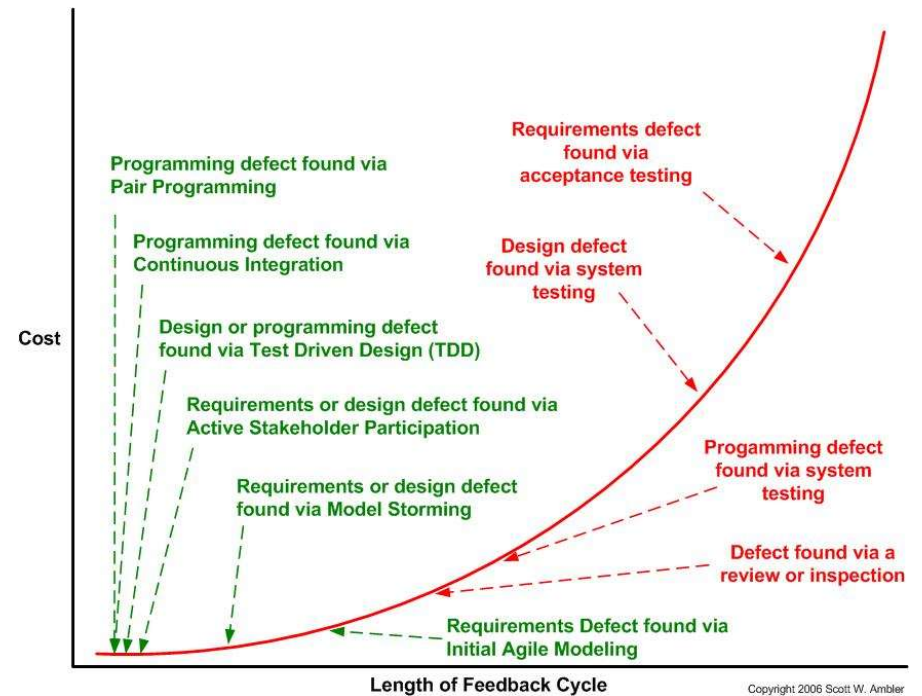
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Defects

- Longer defects are left, more expensive to fix
- More work may have been built on top of bad design, resulting in more work to be undone
- Later in development cycle, more stakeholders impacted by defect and more expensive to fix
- Escaped Defects
 - Defects that make it to the customer



Cost of defects





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Kanban/Task Board

- An "information radiator" - ensures efficient diffusion of information
- Can be drawn on a whiteboard or even a section of wall
- Makes iteration backlog visible
- Serves as a focal point for the daily meeting

Items	In Progress	Testing	Done
	 6 cards	 4 cards	



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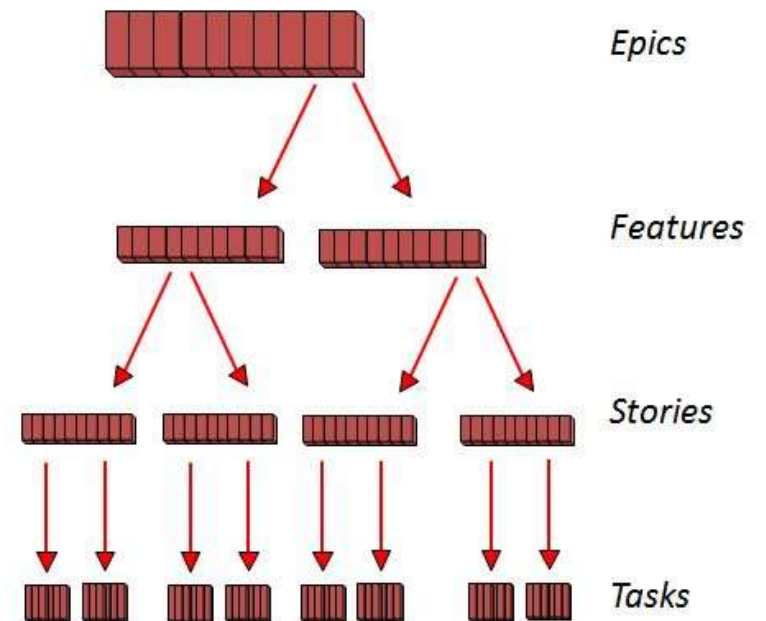
Limit WIP (Work in Progress)

- Includes work that has been started but not completed yet
- Represents money spent with no return
- Hides process bottlenecks that slow the processes
- Represents risk in form of potential risk
- Agile processes aim to Limit and optimize WIP
- Optimal WIP makes processes efficient



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Decomposing Requirements





- User Stories / Backlogs
 - Business functionality within a feature that involves 1-3 days of work.
 - Acts as agreement between customers and development team
 - Every requirement is user story

As a <user type>
I <want to/need, etc.> goal
So that <**value**>



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User Story Example

- “As an payroll clerk, I want to be able to view a report of all payroll taxes, so that I can pay them on time”
- “As a sales person, I want to be able to see a current list of leads, so that I can call them back quickly”
- “As student of this course, I want to be able to understand the requirements of the exam, so that I know if I qualify for it or not”



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Three C's of Stories

- Have users write the stories on index cards
- No details, it's used to help converse
- 3 Cs:
 - Card
 - Conversation
 - Confirmation



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User Stories - INVEST

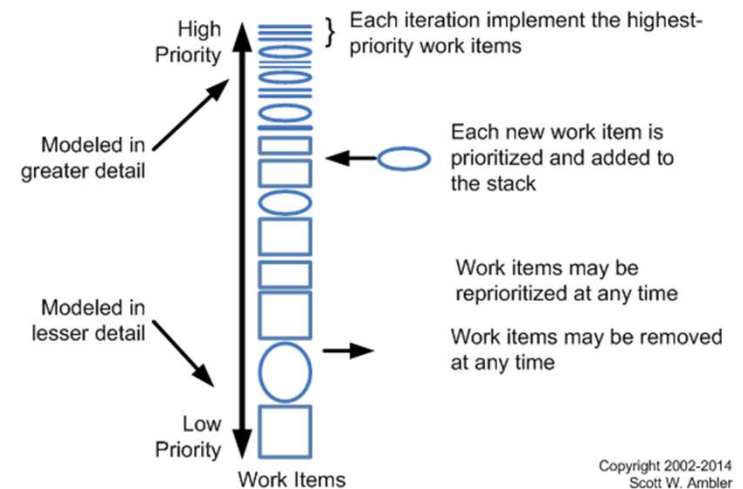
- Effective user stories should be “INVEST”
- **I**ndependent
 - Should be independent so it can reprioritize
- **N**egotiable
 - Should allow for trade-off's based on cost and function
- **V**aluable
 - Should clearly state the value of it
- **E**stimatable
 - Should be able to estimate how long to complete
- **S**mall
 - Stories should be between 4-40 hours of work
- **T**estable
 - Should be testable to ensure it will be accepted once completed



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User Story Backlog (Product Backlog)

- Prioritize Requirements
- Refining (Grooming) Backlog
 - Keeping the backlog updated and accurately prioritized

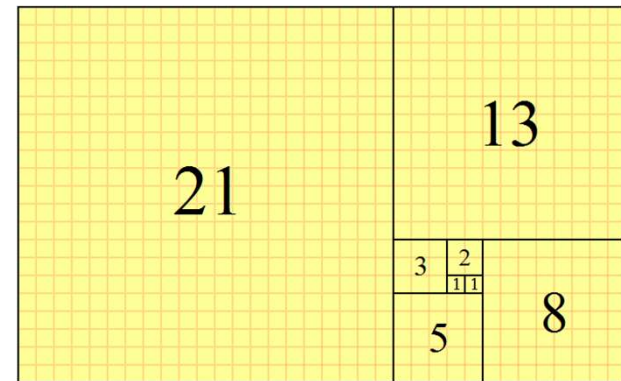


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Fibonacci Sequence



https://en.wikipedia.org/wiki/Fibonacci_number

- Fibonacci Sequence: 1, 2, 3, 5, 8, 13, 21



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Guidelines for Using Story Points

Team should own the definition of their story points

Story point estimates should be all-inclusive

Point sizes should be relative

Complexity, work effort, and risk should all be included in the estimate



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Affinity Estimating and T-Shirt Sizing

- Affinity Estimating
 - Group estimates into categories or collections
- T-Shirt Sizing
 - Place stories in sizes of t-shirts



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Wideband Delphi

- Wideband Delphi
 - Group-based estimation approach
 - Panel of experts, anonymously
- It's used to prevent:
 - Bandwagon effect
 - HIPPO decision making (Highest-Paid Person's Opinion)
 - Groupthink



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Planning Poker

- Advantages of Wideband Delphi
- Fast, collaborative process
- Uses cards with Fibonacci sequence



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Collaboration Games

- Remember the future
- Prune the product tree
- Speedboat(Sailboat)



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Remember the future

Ask stakeholders to imagine that
an upcoming release was
successfully and to look back

Gets a better understanding of
how a stakeholder would define
success

Outlines how we can accomplish
that success for them



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Prune the Product Tree

- Draw a tree and ask stakeholders to add their features to it
- Use stick notes to have them place new features on the tree
- Group the features on the trunk
- Features that are depending on other features would be higher up the tree
- Lets everyone understand the priorities of development



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Speedboat(Sailboat)

- Draw a waterline and a boat moving
- Explain the boat is moving toward the goals of the project
- Ask them to use sticky notes to show what can make the boat move (wind) and what can stop it (anchors)
- Allows stakeholders to identify threats and opportunities



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How Customers Conduct Value Prioritization

- Valued based prioritization is the one of core practices in agile planning
- Features are prioritized on the basis of business value, risk and dependencies
- Some of prioritization techniques used:
 - Simple Scheme
 - MoSCoW prioritization
 - Monopoly Money
 - 100-point method
 - Dot Voting or Multi-voting
 - Kano Analysis
 - Requirements Prioritization Model



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Prioritization Techniques

- Simple Scheme
 - Priority 1, Priority 2, Priority 3, etc.
 - Could be problematic as many items might become the first priority.
- MoSCoW prioritization
 - Must have
 - Should have
 - Could have
 - Would like to have, but not this time



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Prioritization Techniques

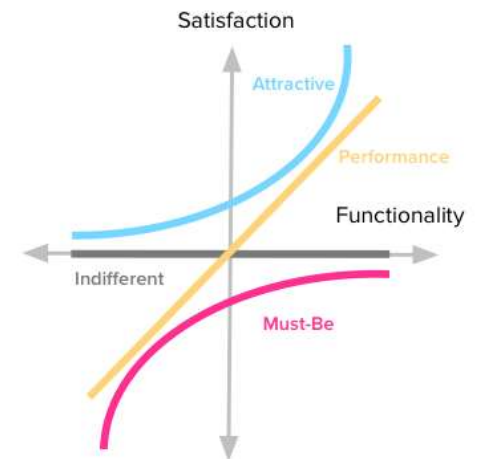
- Dot Voting or Multi-voting
 - Each person gets a certain number of dots to distribute to the requirements
- Monopoly Money
 - Give everyone equal monopoly money
 - They then distribute the funds to what they value the most
- 100-point method
 - Each person is given 100 points
 - They then use that to distribute to individual requirements



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Prioritization Techniques

- Kano Analysis
 - Helps to understand the customers satisfaction
 - Delight
 - Satisfy
 - Dissatisfy
 - Indifferent



<https://foldingburritos.com/kano-model/>



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Prioritization / Ranking is Relative

- Doesn't matter what techniques the customers uses priority, the end results should be a list of prioritized features.



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Lead Time and Cycle Time

- Lead/Cycle time
 - Lead time: how long something takes to go through the entire process
 - Cycle time: how long something takes to go through a part of the process. Part of lead time.
- Cycle Time
 - Measure of how long it takes to get things done
 - Closely related to work in progress (WIP)
 - Excessive WIP is associated with several problems
 - Represents money invested with no return on investment yet
 - Hides bottlenecks in processes & masks efficiency issues
 - Represents risk in form of potential rework



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Little's Law

Helps Agile teams understand flow by connecting three key variables:

- **WIP (Work in Progress)**
- **Throughput (items completed per time period)**
- **Cycle Time (time to complete an item)**

Formula:

- 1. $WIP = Throughput \times Cycle\ Time$**
→ Used when you know how fast work is completed and how long it takes.
- 2. $Cycle\ Time = WIP / Throughput$**
→ Used to calculate how long it takes to complete work.
- 3. $Throughput = WIP / Cycle\ Time$**
→ Used to find out how many items are completed per time period.

Know how to rearrange the formula depending on what's being asked.

It helps ensure predictability and balance in Agile systems.



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Little's Law

Example 1 – Reading a Book (Simple Flow)

You have:

- **WIP:** 60 pages (total work to do)
- **Throughput:** 1 page/day

Question: How long will it take to finish the book?

- **Solution (Cycle Time = WIP / Throughput):**
→ **Cycle Time = 60 / 1 = 60 days**



Little's Law

Example 2 – Forecasting Delivery Time

An Agile team has:

- **WIP:** 10 user stories
- **Throughput:** 5 stories/week

Question: How long will it take to finish the 10 stories?

- **Solution (Cycle Time = WIP / Throughput):**
→ **Cycle Time = $10 / 5 = 2$ weeks**



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Little's Law

Example 3 – Improving Flow Efficiency

Team observes:

- **WIP:** 12 tasks
- **Cycle Time:** 6 days

Question: What is the team's throughput?

- **Solution (Throughput = WIP / Cycle Time):**
→ **Throughput = 12 / 6 = 2 tasks/day**



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Key Performance Indicators (KPI's)

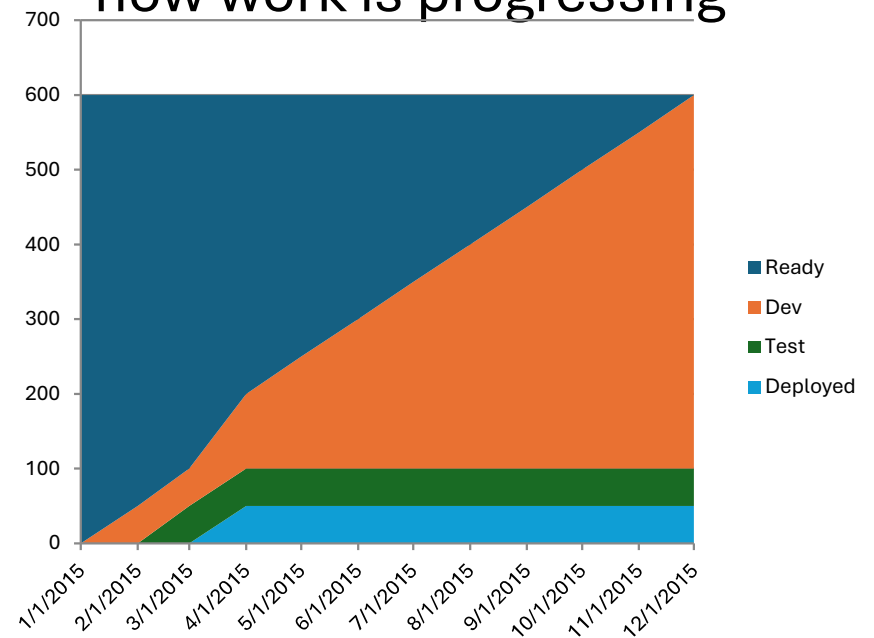
- Uses as a way to measure the project progress
 - Rate of progress: How much points has been completed
 - Remaining work: How much work is yet to be done from the backlog
 - Likely completion date
 - Likely Cost remaining



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Cumulative Flow Diagrams (CFD's)

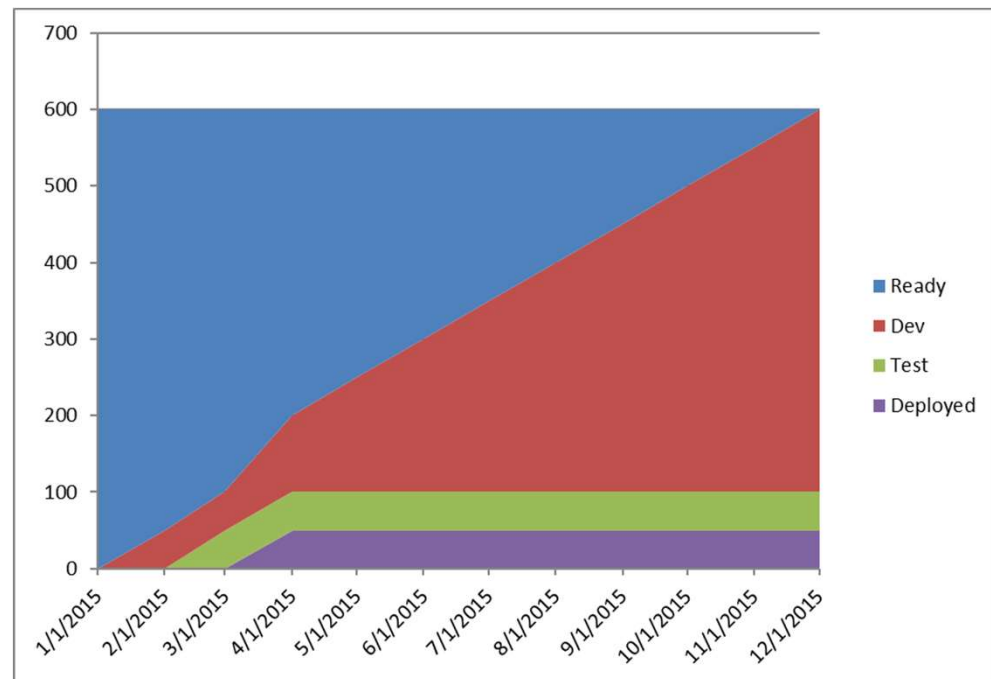
- Stack graphs that show how work is progressing





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Cumulative Flow Diagrams (CFD's)

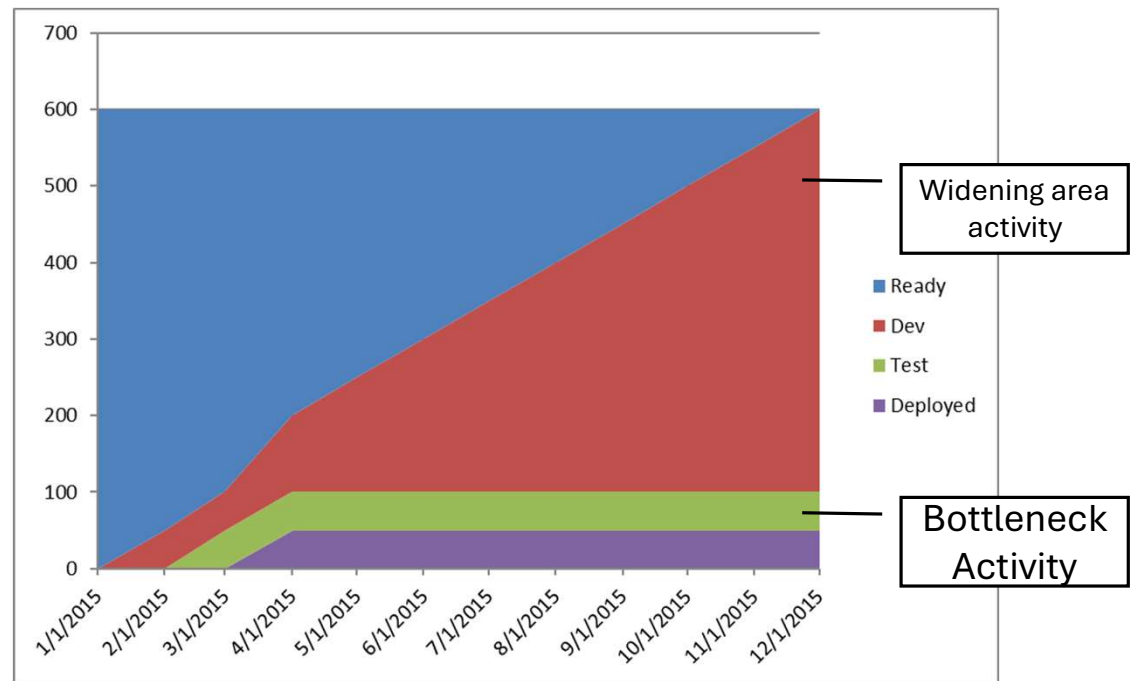


- Bottlenecks and Theory of Constraints



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Cumulative Flow Diagrams (CFD's)



- Bottlenecks and Theory of Constraints



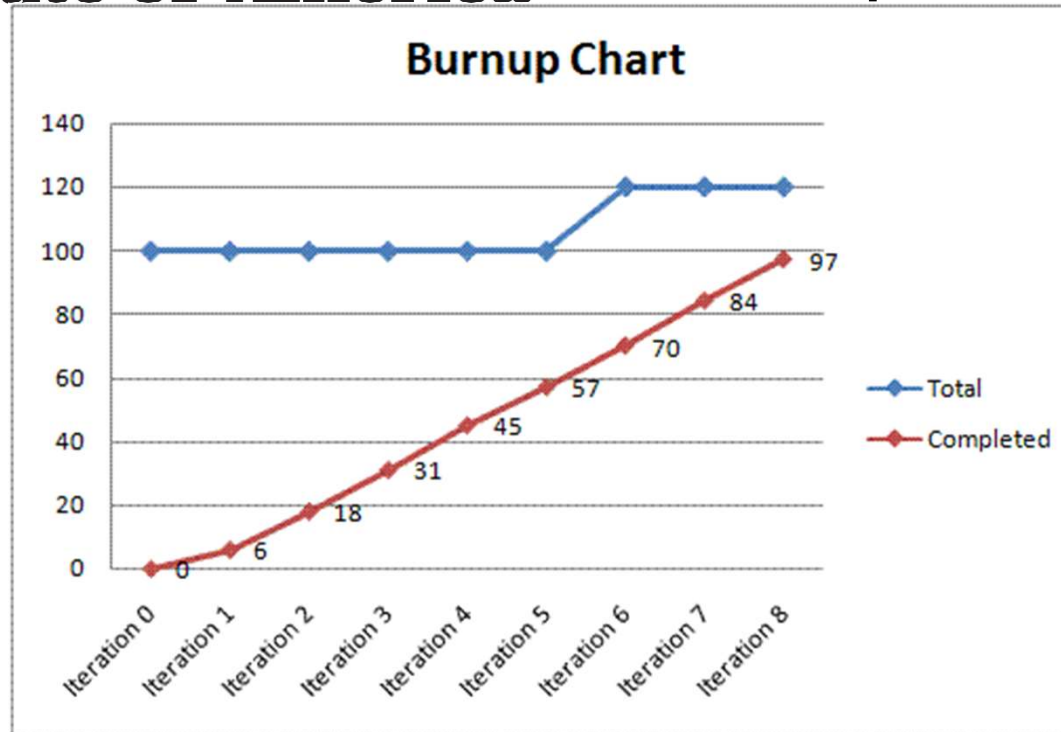
Technical Tracking Team Performance

- Burn Charts
 - Burnup
 - Burndown
- Velocity Charts



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Burnup Chart

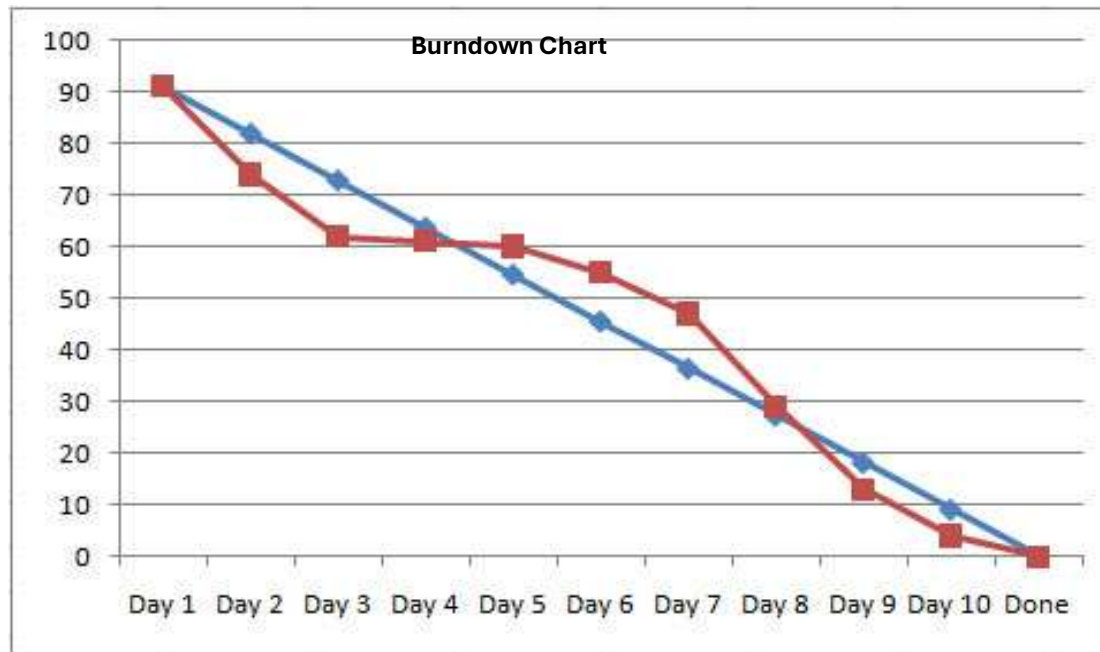


Work that has been done



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Burndown Chart

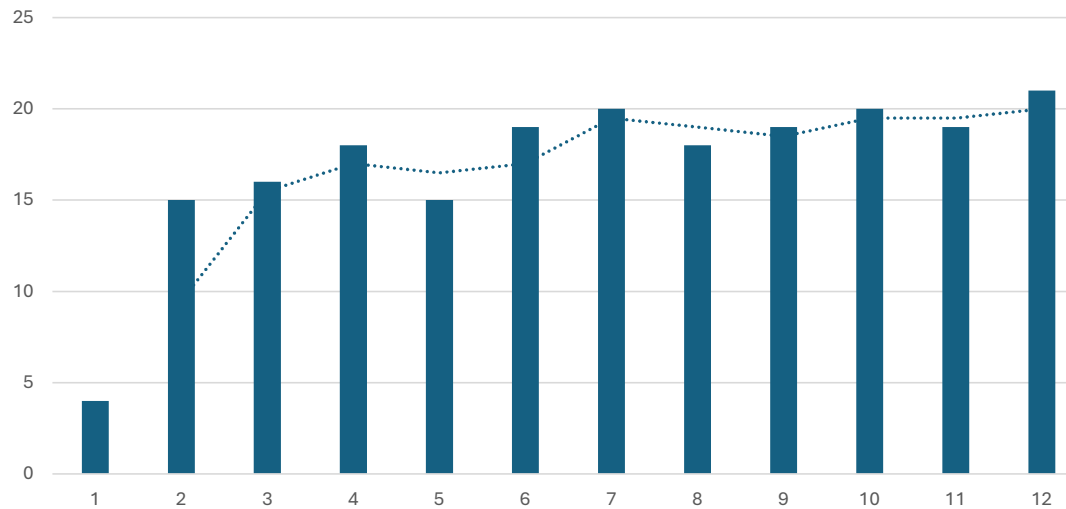


Work that remains to be done



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Velocity Charts



Iterations
Show how the team is performing



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Velocity Charts

- If a team has complete 3 iterations with the average velocity of 18 points per iteration, how many iterations would it take to complete 250 points of work?
- $=250/18 =$ About 14 more iterations.



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Technical Debt

- Backlog of work caused by not doing regular cleanup
- If not done will lead the increase cost of development and make it harder to implement changes
- Refactoring is the solution



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Timeboxing

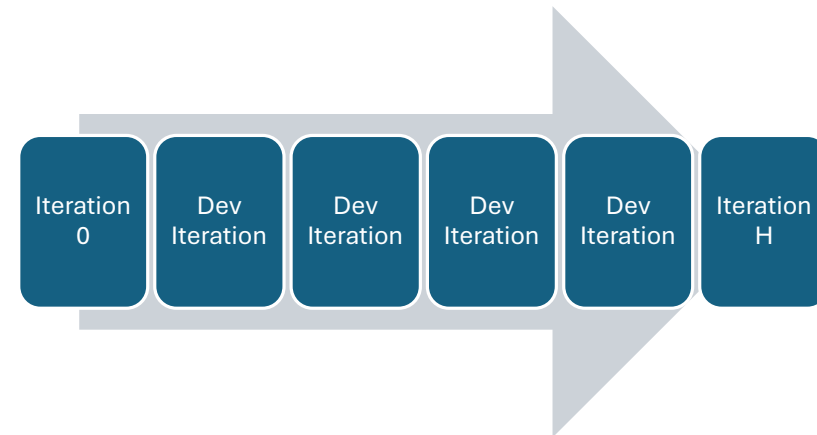
- Short, fixed-duration periods of time in which activities or work are undertaken
 - If work is not completed within time period, move it to another timebox
- Daily Stand-up – 15 minutes
- Retrospectives – 2 hours
- Sprints – 1-4 weeks
- Beware of Parkinson's Law
 - Work tends to expand to fill the time given



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Types of Iterations

- Iteration 0
 - Set the stage for development efforts
 - Doesn't build anything
- Development Iteration
 - Build the product increment
- Iteration H (hardening sprint or release)
 - Done at the end to clean up codes or producing documentation





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Spikes

- Architectural spike
 - Period of time dedicated to proof of concept
- Risk-Based Spike
 - Team investigate to reduce or eliminate risk



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Variance and Trend Analysis

Variance measure of how far apart things are (or vary)

- A measure of how far apart values are from the average.
- Used to track deviations from estimated effort, cost, or timeline.
- Example: If Story Points for sprints vary wildly, variance can highlight inconsistencies in estimation or performance.

Trend Analysis measure that provides insight into future issues

- valuating data over time to forecast future issues or outcomes.
- Common in burnup/burndown charts or velocity trends to predict sprint progress or release readiness.
- Example: A downward velocity trend may indicate team burnout or blockers.
- **Lagging Metrics:** Metrics that reflect what has already occurred.
 - Examples: Delivered story points, number of defects found, actual cost
- **Leading Metrics:** Indicators of what is currently happening or about to happen. Helps anticipate problems before they occur.
 - Examples: Number of tasks in progress, blocked items, WIP (work-in-progress) limits being exceeded.



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Risk Management

- "An uncertain event or condition that, if it occurs, has a positive or negative effect on one or more project objectives."
-PMI
- A risk is something that might happen. If it does happen, it can either help (positive risk, or opportunity) or hurt (negative risk, or threat) the project's:
 - Scope
 - Schedule
 - Cost
 - Quality
 - other objectives
- Considered as anti-value
- Usually has the potential to remove, erode or reduce value with threats



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Managing Risks Process



Traditional Risk Management Approach



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Risk Process

- **Plan Risk Management** – Define how risk management activities will be structured and carried out for the project.
- **Identify Risk** – Proactively determine potential risks that could affect the project and document their characteristics. **Creates a risk register.**
- **Qualitative Risk Analysis** – Prioritize risks based on their probability of occurrence and impact using subjective assessment.
- **Quantitative Risk Analysis** – Numerically evaluate the effect of identified risks on overall project objectives.



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Risk Process

- **Plan Risk Response** – Develop strategies and actions to address high-priority risks and enhance opportunities.
- **Implement Risk Response** – Put the planned risk responses into action during project execution.
- **Monitor and Control Risk** – Continuously track identified risks, reassess existing risks, and evaluate the effectiveness of risk responses.



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Tools to Manage Risk

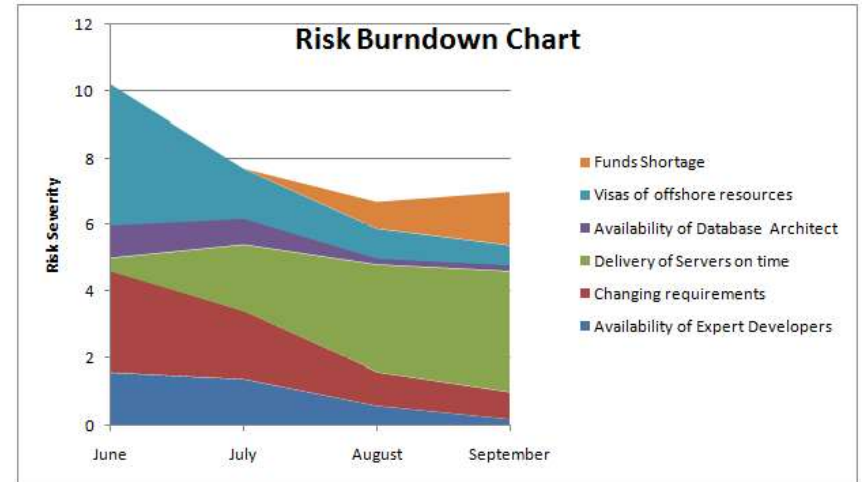
- Risk-adjusted backlog
 - Adjusting the backlog for risk
 - Done after risk response
- Risk spike
 - a **risk spike** (also called a **spike** for risk) is a **time-boxed research activity** used to **explore and reduce uncertainty** around a high-risk or unknown aspect of a project — such as a new technology, integration, or requirement.



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Tools to Manage Risk

- Risk Burndown Graphs





Technical Institute of America Agile Contracting

- Agile's flexibility creates difficulty in outlining contract acceptance criteria
 - Agile attempts to fix resources and time (cost) and vary functionality
- “Customer collaboration over contract negotiation”
 - Close cooperation
 - Active participation
 - Timely and often feedback
- Money for nothing and change for free



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Agile Contracting

- Graduated Fixed Price Contract
 - Buyer / Seller share in risks and rewards
 - Different hourly rates based on:
 - Finish early, Finish on time, Finish late
- Fixed Price Work Packages
 - Mitigate risks of under/over estimating



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CAS (Complex Adaptive Systems)

- A *Complex Adaptive System (CAS)* is a system made up of many individual agents that interact with each other, adapt to change, and evolve over time. In Agile, projects and teams are often treated as CAS.





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CAS (Complex Adaptive Systems)

Characteristics:

- **Emergence:** Patterns and solutions *emerge* from interactions, not top-down planning.
- **Adaptability:** Teams adjust behavior based on feedback and new info.
- **Non-linearity:** Small changes can lead to big, unpredictable results.
- **Self-Organization:** No single point of control – teams organize themselves.
- **Feedback Loops:** Constant feedback shapes team behavior and outcomes.



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CAS (Complex Adaptive Systems)

Agile Relevance:

- Agile frameworks (like Scrum, Kanban, SAFe) embrace CAS thinking.
- Emphasizes experimentation, iterative learning, and responding to change.
- Encourages decentralized decision-making and collaboration.
- **Example:**
Think of an Agile team adapting to market feedback during a sprint – they inspect, learn, and evolve the product and their process.



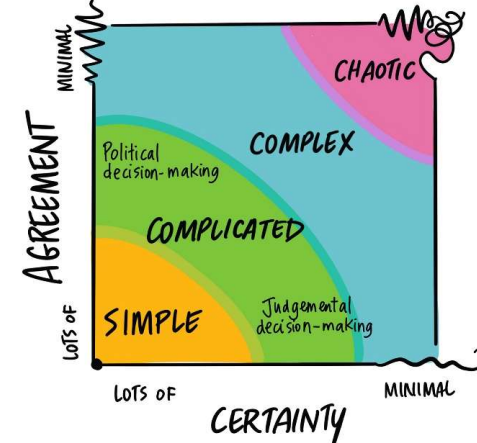
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Stacey Matrix

- A tool used to determine the best management and decision-making approach based on the *level of certainty* and *level of agreement* in a project.

Axes:

- **X-axis**
(Requirements/Agreement):
From *clear* to





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Stacey Matrix

Quadrants:

1. Simple (Clear & Certain):

Best practice works here.
Traditional/waterfall methods are fine.

2. Complicated (Clear but Less Certain):

Expert analysis needed.
Predictive planning can still work.

3. Complex (Unclear Requirements or Uncertain Technology):

Agile fits best.
Use experimentation, feedback loops, and
iterations.

4. Chaotic (No Agreement & No Certainty):

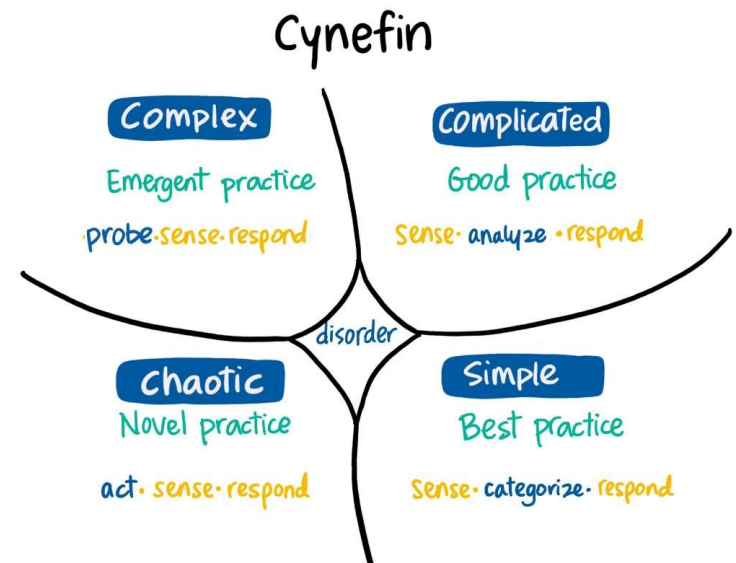
Act first, then sense and respond.
Focus on stabilizing quickly.



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Cynefin

- **Framework** (pronounced *ku-nev-in*) helps leaders make decisions by



Cynefin by Dave Snowden
Frameworks Collection by finegood@sfu.ca | Illustrated by sam@drawingchange.com | © CC BY-NC-ND



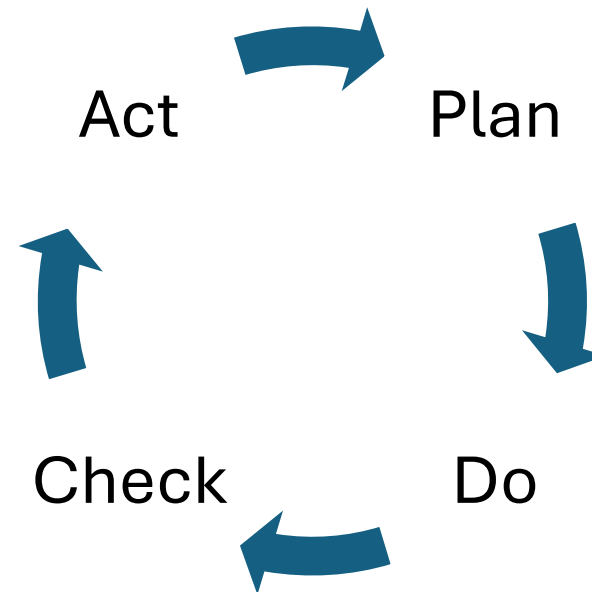
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Domain 4: Delivery



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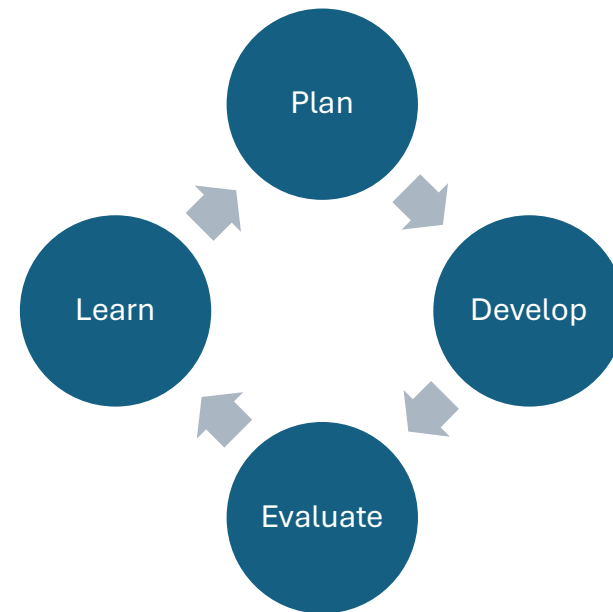
PDCA





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Agile Cycle





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Process Analysis

- Review and diagnose issues
- Look for tailoring possibilities



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Process Tailoring

- Amend methodology to better fit project environment
- Change things for good reason, not just for sake of change
- Develop a hybrid



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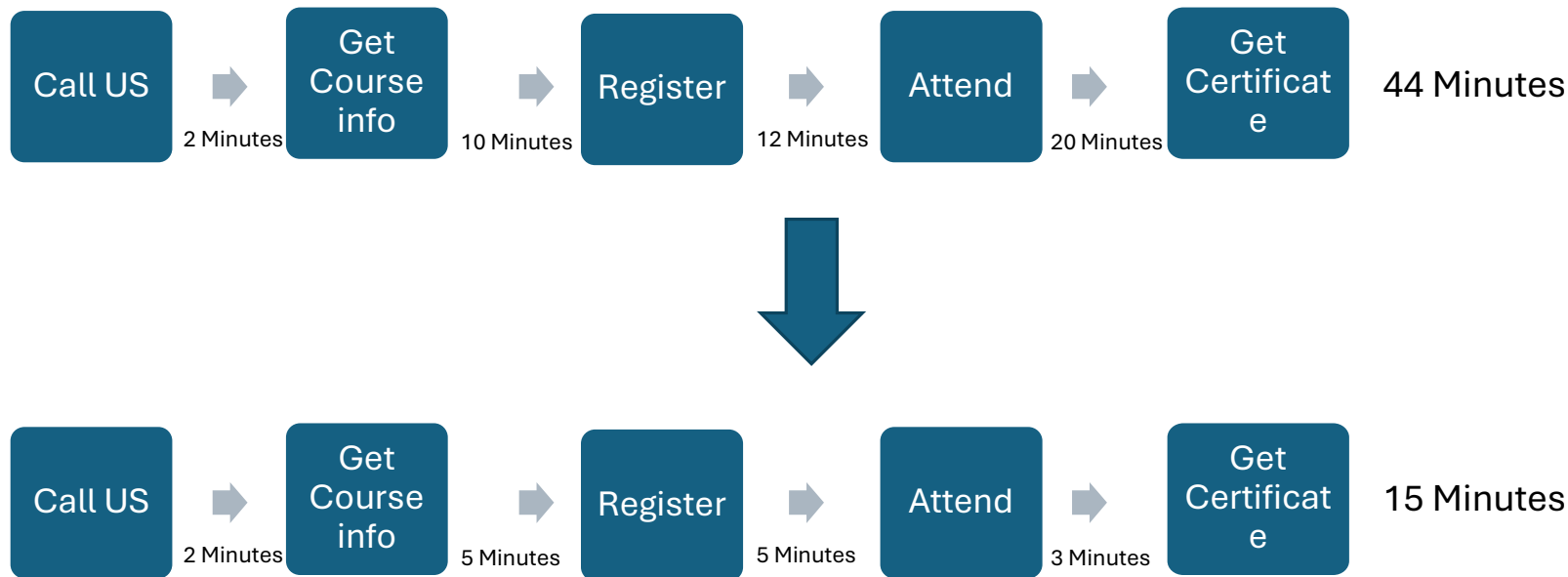
Value Stream Map

- Optimize the flow of information or materials to complete a process
- Reduce waste (waiting times) or unnecessary work
- Steps to creating:
 - Identify the product or service
 - Create a value stream map
 - Review to find waste
 - Create a new map with the desired improvement
 - Develop a roadmap to implement the fixes
 - Plan to revisit it again



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Value Stream Map Example





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Kaizen

- Kaizen is a process for continuous improvement name after the Japanese word
- Focus on the team to implement small incremental improvement
- Usually follows the Plan-Do-Check-Act (PDCA) cycle by Edwards Deming



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Pre-Mortems

- Team meeting that looks at possible things that can cause failure during a project before they take place
- Steps include:
 - Think what the failures might be
 - Create a list of reasons that can cause the failures
 - Review the project plan to determine what can be done to reduce or remove the reasons for failure



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Retrospectives

- Special meeting that takes place after each iteration
- Inspect and improve methods and team work
- Offers immediate value
- Should have a 2 hour time limit



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Retrospectives Stages

About 2 Hours for a typical retrospective

1. Set Stage – 6 Minutes
2. Gather Data – 40 Minutes
3. Generate Insights – 25 Minutes
4. Decide What to Do – 20 Minutes
5. Close Retrospective – 20 Minutes



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1. Set the Stage

- Start of the retrospective
- Help people to get focus
- Encourage participation to ensure everyone start talking early
- Outlining the approach and topics for discussion
- Get people in mood for contributing information
- Activities include:
 - Check-In
 - Focus On/Focus Off
 - ESVP
 - People identify if they are an explorer, shopper, vacationer, or Prisoner



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2. Gather Data

- Create a picture of what happened during the sprint
- Start to collect information to be used for improvement
- Activities:
 - Timeline
 - Triple Nickels: break the team into 5 groups to spend 5 minutes collecting 5 ideas, 5 time
 - Mad, Sad, Glad: what where the team emotion as the sprint was taking place



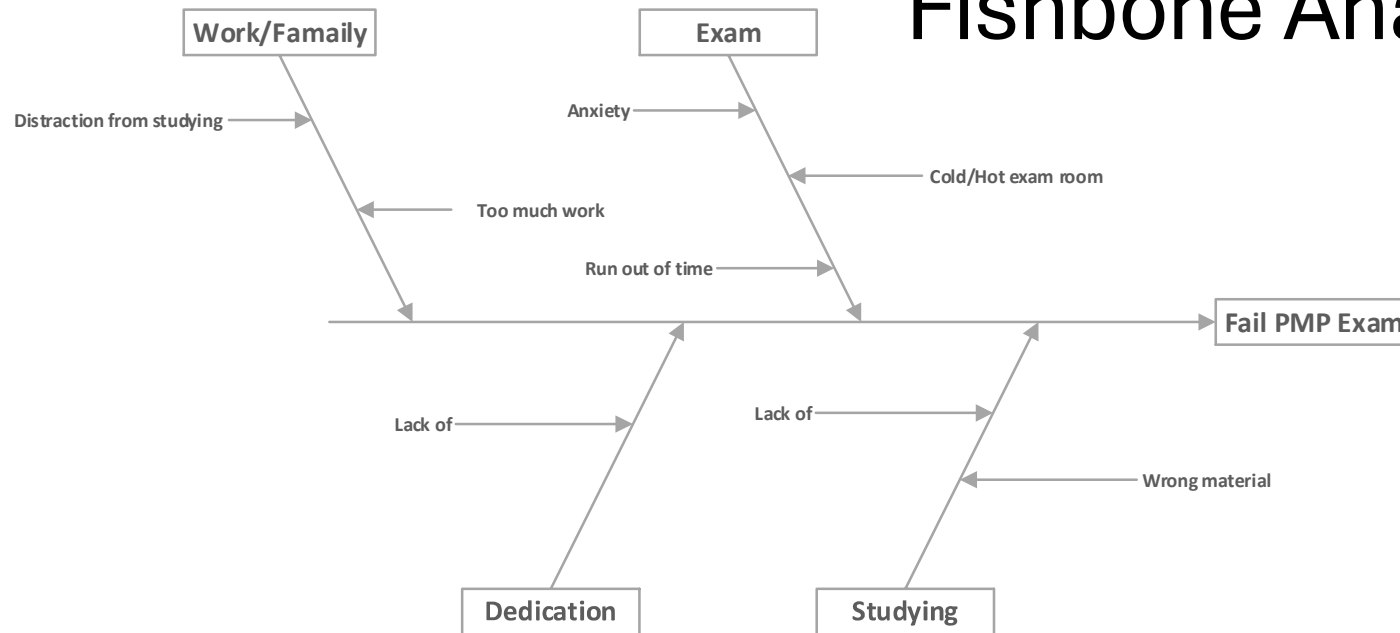
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3. Generate Insights

- Analyze the data
- Helps to understand what was found
- Activities Include:
 - Brainstorming
 - Five Whys: asking why five times
 - Fishbone analysis
 - Prioritize with dots: use a dot voting technique



Fishbone Analysis





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4. Decide what to do

- Decide what to do about the problems that was found
- How can we improve for the next iteration
- Activates include:
 - Short Subjects
 - Smart Goals



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Short Subjects

- Team decides what actions to take in the next iteration:
 - Start doing
 - Stop doing
 - Do more of
 - Do less of



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SMART Goals

- Team sets goals that are SMART:
 - **S**pecific
 - **M**easurable
 - **A**ttainable
 - **R**elevant
 - **T**imely



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The Agile PM Mindset Part 1

- Be a servant leader to the team at all times. This includes empowering them and removing any impediments. Give them the tools they need to succeed while staying out of their way.
- Engage the product owner to document the features and to prioritize them in the product backlog.
- Only the product owner can prioritize the features in the product backlog. If the product owner refuses to do so because they feel all of them are valuable, then you must train them on the benefits of doing so. DO NOT prioritize the features yourself, this is the job of the product owner.



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The Agile PM Mindset Part 2

- Use a co-location
- Face-to-face communications with a white board and markers are the best form of communications.
- Provide agile teams with lots wall space so they can write on them and use sticky notes.



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The Agile PM Mindset Part 3

- Information should always be radiated through the use of large charts and graphs, such as the use of a burnup or burned down chart.
- Any problem that occurs on a project should be resolved by the project team. Always let the project team choose a solution while coaching and supporting their solutions.
- Provide a safe environment for disagreements. Do not punish anyone for having a difference of opinion. Understand that conflicts is a positive step and an opportunity to learn.



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The Agile PM Mindset Part 4

- Try to limit the work in progress through the use of the Kanban. Kanban boards should be displayed either on a large whiteboard or less desirable large monitor.
- Consistently communicate and re-communicate the project vision to the team.
- Understand the needs of your team members and find out what may motivate them



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The Agile PM Mindset Part 5

- Make sure people understand what failure and success will look like on the project.
- Be a central figure to the team, not a dictator.
- Have good ethical values



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The Agile PM Mindset Part 6

- Review the methods work was completed by doing a retrospective
- Utilize feedback loops. Feedback loops occurs when you've completed the task and then take what you've learned from that and input the lessons learned into your next task.